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DESIGN AND IMPLEMENTATION OF AN INTERNET OF THINGS BASED SOLAR GENERATOR FOR NETWORKING DEVICE WITH REGULATED DC-DC OUTPUT OF 5V, 9V, AND 12V

BY

ABDULRAHMAN KIFAYATULLAHI AKOREDE

HNDNCC/24/016

A PROJECT SUBMITTED TO THE DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY, FEDERAL COLLEGE OF ANIMAL HEALTH AND PRODUCTION TECHNOLOGY, MOOR PLANTATION, IBADAN.

IN PARTIAL FULFILLMENT OF REQUIREMENTS FOR THE AWARD OF HIGHER NATIONAL DIPLOMA IN COMPUTER SCIENCE (NETWORKING AND CLOUD COMPUTING)

OCTOMBER, 2026

CERTIFICATION

This is to certify that ***ABDULRAHMAN KIFAYATULLAHI AKOREDE*** with Matriculation Number ***HNDNCC/24/016*** carried out this project at the Department of Computer Science Technology, Federal College of Animal Health and Production Technology, Moor Plantation, Ibadan under my supervisor.

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Mr Ayobioloja, S. P.

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DEDICATION

This project is dedicated to Almighty God.

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ABSTRACT

The increasing dependence on networking devices such as routers, switches, modems, optical network terminals (ONTs), and wireless access points has created a need for reliable and uninterrupted power supply, particularly in areas affected by frequent power outages. Conventional backup power systems may involve additional power conversion stages and often lack real-time monitoring capabilities. This project therefore focused on the design and implementation of an IoT-based solar generator for networking devices with regulated 5V, 9V, and 12V DC outputs. The system integrates a photovoltaic solar panel, solar charge controller, rechargeable battery, DC-DC buck converters, ESP32 microcontroller, voltage and current sensors, and the Blynk IoT platform. Solar energy is converted to electrical energy and stored in the battery, while the DC-DC converters regulate the stored energy to provide the required output voltage levels for compatible networking devices. The ESP32-based monitoring subsystem measures parameters such as battery voltage, output voltage, load current, and power consumption and transmits the data through Wi-Fi to the Blynk cloud dashboard for remote monitoring. System testing showed that the developed unit successfully charged the battery, supplied regulated DC outputs, powered compatible networking loads, and transmitted monitoring data to the Blynk platform. Under tested load conditions, the 5V, 9V, and 12V outputs maintained values close to their respective nominal voltages, with measured efficiencies ranging from 91.8% to 94.2%. The results demonstrate that the developed system provides a functional renewable-energy backup solution capable of supplying regulated DC power to networking devices while enabling real-time remote monitoring of system performance. The project therefore provides an efficient and intelligent approach to improving power availability and continuity for low-voltage networking equipment.

Keywords: Solar power system, Internet of things(IoT), Network devices, Voltage regulator, Battery monitoring, DC power supply, Arduino based monitoring system.

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CHAPTER ONE

INTRODUCTION

1.1 BACKGROUND TO THE STUDY

The continuous operation of digital and network infrastructure constitutes a foundational requirement in the modern technological landscape. Network devices, encompassing routers, switches, modems, and access points, serve as the critical backbone for cloud computing, data transmission, and enterprise connectivity. These devices strictly require a stable, uninterrupted power supply to maintain network availability and prevent catastrophic data loss. However, unreliable grid electricity remains a pervasive challenge in developing regions, frequently disrupting network services and causing significant operational and economic setbacks. To mitigate these disruptions, alternative renewable energy sources are required, with solar energy standing out as a clean, sustainable, and highly reliable solution. Solar power systems convert solar irradiance into electrical energy, which is subsequently stored in battery banks to provide sustained backup power during grid outages (Rezk al., 2021).

Most networking hardware natively operates on Direct Current (DC) at specific low-voltage levels, typically 5V, 9V, or 12V. Providing regulated DC outputs directly from the solar generator avoids unnecessary conversion stages and reduces energy losses. This approach maximizes battery efficiency and significantly extends the operational uptime of connected network devices.

Beyond the physical generation and regulation of power, the dynamic management and monitoring of solar systems are critical to their overall reliability. Traditional solar backup systems largely lack real-time visibility, leaving network administrators unaware of battery degradation, output voltage instability, or impending power failures until the system abruptly shuts down. The integration of the Internet of Things (IoT) systematically addresses this limitation by bridging

physical hardware with cloud-based networks. IoT technology facilitates the continuous acquisition and transmission of system telemetry such as battery voltage, load current, and environmental conditions to a remote dashboard without requiring human intervention (Aliu et al., 2023; Yuan et al., 2025).

111 In furtherance to this study, implementing an IoT-based monitoring system ensures that the solar generator operates safely within defined electrical parameters. Microcontrollers, such as the ESP32 or NodeMCU, act as the central processing unit of the monitoring architecture, reading analog data from voltage and current sensors and transmitting this information over a wireless network to a centralized cloud server (Edeoghon et al., 2022). This infrastructure allows network administrators to remotely monitor the power status of their network devices in real-time. If an anomaly occurs, such as a severe voltage drop or an excessive power draw, the IoT system provides immediate analytical insights, enabling proactive maintenance and preventing sudden network downtime. Consequently, merging regulated DC power distribution with IoT-based cloud monitoring creates a highly efficient, intelligent, and modernized power solution explicitly optimized for networking and cloud computing environments.

1.2 STATEMENT OF THE PROBLEM

Critical network devices require continuous uptime to facilitate seamless data routing and cloud access, yet frequent power outages heavily disrupt this connectivity. Currently, organizations and individuals rely heavily on standard Uninterruptible Power Supplies (UPS) and other conventional backup systems to keep their networking hardware operational during grid failures. However, these existing power systems present significant operational limitations.

Conventional backup systems may require additional power conversion stages before supplying networking devices with the specific DC voltage levels they require. These additional stages can introduce energy losses and reduce the effective operating time of the backup system.

Secondly, traditional power backup systems severely lack intelligent monitoring capabilities. Users typically have no real-time visibility into the exact battery state of charge, the precise amount of load being drawn, or the overall health of the energy system. Without this critical data, a network administrator cannot accurately predict when the backup power will deplete, often resulting in abrupt power failures that corrupt network configurations and prematurely terminate active cloud sessions.

The proposed system resolves these identified lapses by directly integrating regulated DC buck converters into the solar generator, providing highly stable 5V, 9V, and 12V outputs. This hardware arrangement eliminates unnecessary external conversion stages for supported network devices and improves energy efficiency. Furthermore, the proposed system incorporates an IoT-based monitoring module that constantly measures voltage, current, and battery parameters, transmitting this telemetry to a cloud dashboard. This intelligent integration allows users to monitor their power supply remotely, ensuring proactive power management and preventing unexpected network infrastructure downtime.

1.3 AIM AND OBJECTIVES OF THE STUDY

Aim

The aim of this project is to design and implement an Internet of things -based solar generator tailored for network devices, featuring regulated DC outputs of 5V, 9V, and 12V.

Objectives

To effectively achieve the aim of this project, the following objectives were outlined:

I. To integrate a solar-powered battery charging subsystem with regulated DC outputs suitable for network devices.

II. To develop a power distribution unit that delivers stable 5V, 9V and 12V DC simultaneously.

III. To implement voltage, current, and battery level sensing mechanisms for real-time power monitoring.

IV. To integrate IoT communication technology for transmitting system performance data to a cloud server.

V. To develop a cloud-based dashboard for remote monitoring of battery status, output voltage levels, and load conditions.

1.4 SIGNIFICANCE OF THE STUDY

The significance of this study lies in its capacity to provide a highly efficient, intelligent, and tailored power architecture for modern digital infrastructure. The primary benefit of this system is the reduction of energy wastage by supplying direct, regulated DC power to network devices while providing real-time monitoring of the power system.

The primary beneficiaries of this project include network administrators, Internet Service Providers (ISPs), remote IT professionals, and small-to-medium enterprises (SMEs). For network administrators and ISPs, the system guarantees that critical networking hardware, such as core switches and access points, remains online during grid outages, thereby preserving seamless cloud connectivity and data communication.

Furthermore, these users will benefit heavily from the integrated IoT framework. The cloud-based dashboard empowers administrators to remotely monitor the battery lifecycle and power

consumption of the generator from any geographical location. This remote visibility significantly reduces manual maintenance costs, prevents unexpected network downtimes, and protects the battery from deep discharge or overcharging, ultimately extending the longevity of the entire power system.

1.5 SCOPE AND LIMITATIONS OF THE STUDY

This project strictly focuses on the design, construction, and computational evaluation of a solar generator equipped with regulated DC buck converters. Functionally, the system is engineered to provide precisely regulated DC outputs fixed at 5V, 9V, and 12V to accommodate various network devices like modems, routers, and edge switches. The scope extends to the programming and integration of an ESP32 microcontroller, interfaced with voltage and current sensors, to push real-time telemetry data to an IoT cloud dashboard (Blynk).

In terms of limitations, the hardware's operational capacity is constrained by the physical capacity of the integrated battery and the wattage of the solar panel utilized in the prototype. This dictates the maximum concurrent load the generator can sustain. Additionally, the remote monitoring functionality is strictly dependent on the availability of an active local wireless internet connection. Should the local Wi-Fi network experience an outage, the microcontroller will be unable to update the cloud dashboard in real-time, although the physical power generation and voltage regulation will continue to function autonomously.

1.6 DEFINITION OF TERMS

Internet of Things (IoT): A network of physical electronic devices embedded with sensors, software, and communication hardware that allows them to autonomously collect and exchange data over the internet.

Solar Generator: A self-contained, portable power architecture that captures irradiance from the sun via photovoltaic panels, stores the electrical energy in a battery, and distributes it to connected loads.

Regulated DC Output: A power supply terminal that utilizes voltage regulation circuitry to maintain a constant, specific Direct Current voltage level (e.g., 5V, 9V, 12V) regardless of input fluctuations from the battery.

Regulated DC Output: A power supply terminal that utilizes buck or boost converters to maintain a constant, specific Direct Current voltage level (e.g., 5V, 9V, 12V) regardless of input fluctuations from the battery.

129 **NodeMCU/ESP32:** A low-cost, open-source microcontroller board featuring built-in Wi-Fi and Bluetooth capabilities, extensively utilized for transmitting telemetry data in IoT applications.

Cloud Dashboard: A web-based graphical user interface that retrieves telemetry data from a cloud server to visually display real-time metrics, such as battery health and system power consumption.

Network Devices: Core IT hardware components, such as routers, switches, and modems, engineered to interconnect computers and facilitate data routing across local and wide-area networks.

CHAPTER TWO

LITERATURE REVIEW

This chapter provides a literature review on designing an IoT-enabled solar generator that delivers regulated 5V, 9V, and 12V DC outputs for networking devices, framing the context around solar energy conversion, battery storage, charge controllers, DC regulation, and the power challenges facing network infrastructure. It evaluates IoT integration focusing on microcontrollers, wireless communication, cloud monitoring, energy management, and system protection while analyzing previous empirical studies to highlight existing research gaps and establish the conceptual framework for the proposed system.

2.1 CONCEPT OF SOLAR GENERATOR SYSTEMS

A solar generator system is a renewable-energy system that converts solar radiation into electrical energy for direct use or storage. Unlike conventional fuel generators, it does not require continuous petrol or diesel consumption and is suitable for areas with unreliable or unavailable grid electricity. Solar PV technology is highly scalable and can support applications ranging from small off-grid systems to large electricity-generation facilities (IEA, 2022). IRENA (2024) also identifies off-grid renewable-energy systems as suitable for telecommunications and other remote facilities.

A typical solar generator comprises PV modules, a solar charge controller, rechargeable battery, protection devices, and power-conversion components such as DC-DC converters or an inverter. The PV modules generate DC electricity, the charge controller manages battery charging, and the battery stores energy for use when solar radiation is unavailable. For networking equipment, DC-

DC converters are important because devices such as routers, switches, modems, and wireless access points commonly require regulated low-voltage DC power (NREL, 2024).

Solar generators provide renewable and relatively low-emission power, but their performance depends on solar availability and environmental conditions (Ahmed, 2022). Battery storage also requires appropriate charging and protection. Studies by Javidsharifi et al. (2021) and Baidas et al. (2022) demonstrate the potential of renewable-energy systems for powering telecommunications infrastructure. Therefore, proper sizing and voltage regulation are essential for reliable operation of solar-powered networking systems. These requirements form the basis of the proposed IoT-enabled solar generator for regulated DC power supply and real-time monitoring of networking devices.

2.2 SOLAR PHOTOVOLTAIC (PV) TECHNOLOGY

Photovoltaic (PV) technology converts sunlight directly into electricity using semiconductor materials. Its scalability makes it suitable for both small standalone systems and large-scale electricity generation, including off-grid applications (IEA, 2020). Continued technological development has also contributed to improved PV performance and deployment (IRENA, 2025).

2.2.1 SOLAR PANELS, PHOTOVOLTAIC CELLS AND CONVERSION OF SOLAR ENERGY INTO ELECTRICAL ENERGY

A solar panel or PV module consists of interconnected photovoltaic cells, commonly manufactured from crystalline silicon. When sunlight strikes the cells, photons transfer energy to electrons in the semiconductor, producing direct-current (DC) electricity. Cells are connected to form modules,

while modules can be combined in series or parallel to obtain the required voltage and current (IEA, 2020).

The generated DC electricity may be supplied directly to DC loads or stored in a battery through a charge controller. In the proposed system, the PV output is used to charge the battery, after which the stored energy is regulated into 5 V, 9 V and 12 V DC outputs suitable for networking devices.

2.2.2 FACTORS AFFECTING SOLAR PANEL PERFORMANCE AND SOLAR PANEL SIZING FOR SMALL ELECTRONIC SYSTEMS

Solar panel performance is influenced by factors such as solar irradiance, cell temperature, shading, orientation, tilt angle, soiling, module technology, degradation, and system losses. Higher temperatures generally reduce PV efficiency, while reduced sunlight and shading lower electrical output. NREL (2021) and IEA (2021) emphasise the importance of these technical and environmental factors in PV performance and energy yield.

Proper panel sizing is therefore necessary to meet the energy requirements of the connected load. For small networking systems, sizing should consider total load power, operating hours, solar availability, battery requirements, system losses, and a design margin. Energy consumption can be estimated using:

$$\mathbf{E = P \times t}$$

where E is energy consumption, P is load power, and t is operating time. The PV array should generate sufficient energy to operate the load and recharge the battery after accounting for system losses (IEA, 2020). This is important for the proposed system because networking devices may

operate continuously, and inadequate PV capacity could result in battery depletion and interruption of service.

2.2.3 APPLICATIONS OF PV SYSTEMS IN OFF-GRID ENVIRONMENTS

PV systems are widely used where grid electricity is unavailable, unreliable, or expensive to extend. Applications include solar home systems, lighting, water pumping, telecommunications, charging stations, schools, clinics, and remote commercial facilities. IRENA (2024) reports continued growth in off-grid renewable-energy deployment, including applications in telecommunications and remote facilities.

The modularity of PV systems and their ability to operate independently of the national grid make them suitable for remote applications. Batteries can also be incorporated to provide electricity when solar generation is unavailable (IEA, 2020; IRENA, 2024). In the proposed system, PV serves as the primary energy source, with generated electricity stored and regulated to provide 5 V, 9 V, and 12 V DC outputs for networking equipment.

2.3 BATTERY ENERGY STORAGE IN SOLAR GENERATOR SYSTEMS

Battery energy storage enables electricity generated by solar PV systems to be stored for use when solar radiation is insufficient. It therefore provides an important link between intermittent solar generation and continuous electrical loads. The IEA (2024) identifies battery storage as an increasingly important technology for integrating variable renewable energy and improving electricity availability.

2.3.1 PURPOSE OF BATTERIES IN SOLAR SYSTEMS

The primary function of a battery in a standalone solar system is to store excess energy generated during periods of adequate sunlight and supply it when solar generation is low or unavailable. This is particularly important for systems that must operate beyond daylight hours. Battery-supported renewable-energy systems are used in telecommunications, rural facilities, charging stations, and other remote applications (IRENA, 2023; IEA, 2024).

For the proposed solar generator, the battery provides stored energy for networking devices during periods of insufficient solar generation or power interruption.

2.3.2 TYPES OF BATTERIES USED IN SOLAR APPLICATIONS

Lead-acid and lithium-ion batteries are commonly used in standalone solar systems. Lead-acid batteries have established technology and relatively low initial cost, while lithium-ion batteries generally offer higher energy density, efficiency, cycle life, and lower weight. Lithium iron phosphate (LFP) has become an important lithium-ion chemistry for stationary storage because of its suitability for frequent cycling. NREL (2024) identifies LFP and NMC lithium-ion batteries among representative stationary-storage technologies, while the IEA (2026) reports the growing dominance of LFP in battery-storage deployments.

Battery selection depends on capacity, depth of discharge, cycle frequency, cost, operating temperature, maintenance, safety, and expected service life. For small solar generators, lithium-based batteries offer compact and efficient storage, while lead-acid batteries may remain suitable where lower initial cost is prioritized.

2.3.3 BATTERY CAPACITY AND ENERGY REQUIREMENTS

Battery capacity indicates the amount of energy that can be stored and is commonly expressed in ampere-hours (Ah) or watt-hours (Wh). Proper sizing depends on the load demand, required backup duration, battery voltage, and allowable depth of discharge, system efficiency, and operating conditions. Nominal battery energy can be estimated as:

$$E = V \times C$$

Where E is battery energy in watt-hours, V is battery voltage, and C is capacity in ampere-hours. The actual usable energy is lower because of allowable depth of discharge and system losses.

For the proposed system, battery capacity should therefore be based on the combined energy requirements of the networking devices and the required backup period. Adequate capacity is essential for maintaining network availability during periods of low solar generation or grid failure.

2.3.4 BATTERY CHARGING AND DISCHARGING

Battery charging involves storing electrical energy from the PV system, while discharging occurs when the battery supplies energy to the load. The charge controller regulates the charging voltage and current to protect the battery. When solar generation exceeds the load demand, excess energy can charge the battery; when solar generation falls below the load requirement, the battery supplies the difference.

Repeated charging and discharging affect battery life. Battery degradation is influenced by factors such as chemistry, depth of discharge, operating temperature, and charging conditions.

Appropriate operating limits are therefore necessary to maintain battery performance and service life.

2.3.5 BATTERY MANAGEMENT AND PROTECTION

Battery management is necessary to maintain safe and reliable operation. Important protection functions include overcharge, over-discharge, overcurrent, short-circuit, and temperature protection, while battery packs with multiple cells may also require cell balancing. A battery management system (BMS) can monitor these parameters and disconnect or regulate the battery when unsafe conditions occur.

Effective battery protection is particularly important for lithium-ion systems because unsuitable charging or discharge conditions can reduce battery life and create safety risks. The proposed system therefore requires appropriate charging, monitoring, and protection mechanisms.

2.3.6 IMPORTANCE OF ENERGY STORAGE FOR NETWORKING DEVICES

Battery storage is important for networking devices because routers, switches, modems, ONTs, and wireless access points may need to operate continuously to maintain communication and Internet connectivity. A solar-battery system can provide backup power during grid interruptions and periods of inadequate solar generation.

For the proposed project, battery storage provides the connection between intermittent solar energy and the continuous power requirements of networking equipment. Adequate storage capacity, charging control, monitoring, and protection are therefore necessary to maintain the regulated 5 V, 9 V, and 12 V DC outputs required by the connected devices.

2.4 SOLAR CHARGE CONTROLLER

A solar charge controller is installed between the PV array and battery to regulate the voltage and current supplied during charging. It prevents overcharging and excessive battery discharge while managing energy transfer between the PV source, battery, and loads. Effective charge control improves battery protection, energy utilization, and system reliability (Tlili et al., 2023).

2.4.1 FUNCTIONS OF A CHARGE CONTROLLER

The main functions of a charge controller include regulating charging voltage and current, preventing overcharging, and disconnecting loads when battery voltage falls below a safe level. Some controllers also provide temperature monitoring, overcurrent protection, and system-status indicators. IEEE Standard 1562-2021 identifies overcharge and undercharge prevention as principal functions of standalone PV charge controllers, with low-voltage load disconnection and overcurrent protection as additional functions (IEEE, 2021).

2.4.2 PWM CHARGE CONTROLLERS

Pulse Width Modulation (PWM) controllers regulate charging by rapidly switching the connection between the PV array and battery. They are relatively simple and inexpensive but do not operate the PV panel at its maximum power point, which can result in lower energy utilization. PWM controllers are therefore more suitable for small and relatively simple solar systems (Schmid & Behrendt, 2021).

2.4.3 MPPT CHARGE CONTROLLERS

Maximum Power Point Tracking (MPPT) controllers use electronic control and DC-DC conversion to operate the PV array near its maximum power point. Because this point changes with irradiance and temperature, MPPT allows more of the available PV energy to be extracted than conventional PWM control (Babu et al., 2020; Ahmed et al., 2020). Although generally more expensive, MPPT is advantageous where efficient energy harvesting is important. It may therefore be considered for the proposed system where maximum utilization of available solar energy is required.

2.4.4 BATTERY OVERCHARGING PROTECTION

Overcharging occurs when a battery continues receiving energy after reaching its appropriate charging level. It can cause degradation, heating, reduced service life, and safety problems. The charge controller prevents this condition by monitoring battery voltage and regulating or interrupting charging when the required level is reached (IEEE, 2021).

2.4.5 DEEP-DISCHARGE PROTECTION

Deep discharge occurs when a battery is discharged below its recommended voltage or state of charge. Repeated deep discharge can reduce battery capacity and service life. Charge controllers can provide low-voltage load disconnection to protect the battery when it reaches a predetermined minimum level (IEEE, 2021). This function is particularly important for networking applications where connected devices may otherwise continuously drain the battery.

2.5 DC POWER SUPPLY AND VOLTAGE REGULATION

A DC power supply provides controlled voltage and current to electronic loads. Since battery voltage varies with charging, discharging, and load conditions, a voltage-regulation stage is required to provide stable outputs. DC-DC converters are commonly used because they can step voltage up or down while maintaining regulated output levels (Nandankar et al., 2021).

2.5.1 IMPORTANCE OF REGULATED DC VOLTAGE

Electronic equipment operates within specified voltage ranges, and voltage variations can cause malfunction, overheating, reduced service life, or shutdown. Since the battery voltage is not constant, direct connection of sensitive networking equipment may provide unsuitable voltage. Voltage regulation therefore provides a stable interface between the battery and connected devices (Bel, 2020).

2.5.2 VOLTAGE REGULATION TECHNIQUES AND LINEAR VOLTAGE REGULATORS

Voltage regulation can be achieved using linear or switching regulators. Linear regulators are simple, inexpensive, and produce relatively low-noise outputs, but they dissipate excess voltage as heat and therefore become inefficient at high current or large voltage differences. For example, reducing 12 V to 5 V with a linear regulator causes the unused voltage to be dissipated as heat. They are consequently more suitable for low-current auxiliary circuits than the main outputs of the proposed system (Bel, 2020).

2.5.3 SWITCHING DC-DC CONVERTERS

Switching DC-DC converters regulate voltage through high-frequency switching and energy storage components such as inductors and capacitors. They generally achieve higher efficiency than linear regulators and are suitable for battery-powered and renewable-energy applications (Nandankar et al., 2021). Common converter types include buck, boost, and buck-boost converters (Ibrahim et al., 2021; Wu et al., 2023).

2.5.3.1 BUCK CONVERTERS

A buck converter reduces a higher input voltage to a lower regulated output voltage. It is therefore suitable for applications such as converting a 12 V battery supply to 9 V or 5 V. Buck converters are widely used because of their relatively simple structure and high efficiency (Nandankar et al., 2021; Wu et al., 2023).

2.5.3.2 BOOST CONVERTERS

A boost converter increases a lower input voltage to a higher output voltage. It is useful where the available battery or auxiliary DC voltage is below the required output level. Boost converters are also commonly used as interfaces between renewable-energy sources and electrical loads (Sudarsan Reddy & Thangavel, 2021).

2.5.3.3 BUCK-BOOST CONVERTERS

A buck-boost converter can increase or decrease the input voltage, making it useful where the source voltage varies above and below the required output level. This characteristic can be

advantageous in solar systems because battery voltage changes during charging and discharging (Wu et al., 2023).

2.5.4 EFFICIENCY AND HEAT DISSIPATION

Efficiency is important because energy lost during voltage conversion reduces the energy available to the load. Linear regulators mainly lose energy as heat, while switching converters generally achieve higher efficiency but still experience switching, conduction, and magnetic losses (Nandankar et al., 2021; Ibrahim et al., 2021). Proper component selection, ventilation, and heat dissipation are therefore necessary for reliable operation.

2.5.5 IMPORTANCE OF STABLE DC VOLTAGE FOR NETWORKING EQUIPMENT

Routers, switches, modems, ONTs, and wireless access points require specified supply-voltage ranges. Unstable voltage can cause malfunction, resets, communication interruptions, or equipment damage. The proposed solar generator therefore uses regulated 5 V, 9 V, and 12 V DC outputs to accommodate different networking devices. DC-DC conversion also avoids unnecessary DC-to-AC-to-DC conversion, thereby reducing potential energy losses (Nandankar et al., 2021; Wu et al., 2023).

2.6 NETWORKING DEVICES AND THEIR POWER REQUIREMENTS

Networking devices provide the infrastructure required to establish and maintain communication between computers, users, servers, sensors, and Internet-connected systems. Common devices include routers, switches, wireless access points, modems, optical network terminals (ONTs), and IoT gateways. Their continuous operation depends on an appropriate and reliable electrical supply.

2.6.1 OVERVIEW OF COMMON NETWORKING DEVICES

Networking devices perform different but complementary functions. Routers connect and direct traffic between networks, switches connect devices within local networks, wireless access points provide Wi-Fi connectivity, modems provide access to telecommunications services, and ONTs interface fiber-optic networks with customer equipment. IoT gateways connect IoT devices to wider networks or cloud platforms. Their power requirements vary according to hardware configuration, processing capacity, communication interfaces, and additional features.

2.6.1.1 ROUTERS

A router connects different networks and forwards data packets using network addressing and routing information. Modern routers may also include wireless, firewall, VPN, and cellular functions, resulting in different power requirements among models. Therefore, the actual voltage and power requirements of the selected router should be considered when designing the solar power system.

2.6.1.2 NETWORK SWITCHES

A network switch connects multiple devices within a local network and forwards Ethernet traffic between ports. PoE-capable switches can also supply power to devices such as wireless access points, IP cameras, and IP phones. Consequently, the total power requirement of a PoE switch includes both its own consumption and the power delivered to connected devices.

2.6.1.3 WIRELESS ACCESS POINTS

A wireless access point provides wireless connectivity between client devices and a wired network. Its power consumption depends on factors such as the number of radios, Wi-Fi technology, processing functions, and PoE capability. Since access points commonly operate continuously, their power requirements must be considered when sizing the solar system.

2.6.1.4 MODEMS

A modem provides the interface between a local network and a telecommunications service such as DSL, cable, or cellular connectivity. Because it normally remains active to maintain Internet access, it represents a continuous electrical load. Its voltage and current requirements depend on the specific model.

2.6.1.5 OPTICAL NETWORK TERMINALS (ONTS)

An Optical Network Terminal (ONT) provides the interface between a fibre-optic access network and the customer's electrical/data network. ONTs are commonly used for fibre Internet connectivity and normally require continuous power for network access. Their power requirements should therefore be considered when designing a solar-powered network supply.

2.6.1.6 IOT GATEWAYS

An IoT gateway provides communication and processing between IoT devices and wider networks or cloud platforms. It can collect sensor data, perform protocol conversion, and transmit information remotely. In the proposed system, an IoT controller can perform similar monitoring functions by collecting information on voltage, current, battery condition, and power availability.

2.6.2 POWER REQUIREMENTS OF NETWORKING DEVICES

The power requirements of networking equipment vary according to device type, manufacturer, configuration, and operating mode. They are commonly specified using input voltage, current, or power ratings. When several devices operate simultaneously, their individual power requirements must be combined to determine the total system load. The solar panel, battery, charge controller, and DC-DC converters must therefore be sized according to the total demand and the specified input voltage of each device.

The proposed 5 V, 9 V, and 12 V DC outputs are intended to accommodate different small networking and electronic devices while avoiding unnecessary conversion between DC and AC.

2.6.3 EFFECTS OF UNSTABLE POWER ON NETWORKING EQUIPMENT

Networking equipment requires power within specified operating limits. Voltage fluctuations, under-voltage, over-voltage, and interruptions can cause malfunction, unexpected restarts, communication interruptions, or shutdown. Loss of power to a router, switch, modem, or access point can therefore immediately interrupt the network services provided by that device.

Maintaining stable power is consequently an important part of ensuring network service continuity.

2.6.4 IMPORTANCE OF UNINTERRUPTED POWER FOR NETWORK AVAILABILITY

Continuous power is essential because network services depend on the operation of routers, switches, access points, modems, and ONTs. Power interruptions can result in loss of Internet access, local-network communication, wireless connectivity, and cloud-based services. Backup

batteries and other power systems are therefore widely used where network availability is important.

For the proposed project, the combination of solar generation, battery storage, regulated 5 V, 9 V, and 12 V DC outputs, and IoT-based monitoring provides an alternative power source for selected networking devices and supports continued operation during grid interruptions or inadequate solar generation.

2.7 POWER CHALLENGES IN NETWORK INFRASTRUCTURE

Reliable electricity is essential for network infrastructure because routers, switches, wireless access points, modems, ONTs, and other communication equipment require continuous power. In Nigeria, unreliable electricity supply remains a significant challenge, with frequent outages and variations in supply quality (World Bank, 2025). These conditions create a need for dependable backup power systems for network equipment.

2.7.1 UNRELIABLE GRID ELECTRICITY

Unreliable grid electricity is characterized by frequent outages, interruptions, and voltage variations. In Nigeria, these challenges have led households, businesses, telecommunications operators, and other organizations to rely on alternative power sources such as petrol and diesel generators. For network infrastructure, interruptions can directly affect Internet connectivity and digital services.

2.7.2 POWER INTERRUPTIONS AND NETWORK DOWNTIME

Power interruptions can cause routers, switches, modems, access points, and ONTs to shut down, resulting in network downtime until power is restored and the equipment restarts. Uptime Institute (2024) identifies power issues as a major cause of serious data-center outages, demonstrating the close relationship between power reliability and continuity of digital services.

2.7.3 DEPENDENCE ON CONVENTIONAL BACKUP GENERATORS

Petrol and diesel generators are widely used as backup sources where grid electricity is unreliable. Although they can provide relatively high-power output and operate independently of solar availability, they require continuous fuel and regular maintenance. Heinemann et al. (2022) identify fuel-powered generators as a common backup option for telecommunications infrastructure while highlighting their economic and environmental disadvantages.

2.7.4 FUEL AND MAINTENANCE COSTS

The operating cost of conventional generators includes fuel, servicing, repairs, and replacement parts. Frequent operation due to repeated grid failures can therefore result in significant long-term costs. Heinemann et al. (2022) note that fuel and maintenance contribute substantially to the lifetime cost of diesel and gasoline generators. This creates an economic incentive to consider renewable-energy alternatives where the load and environmental conditions are suitable.

2.7.5 POWER FLUCTUATIONS AND EQUIPMENT DAMAGE

Power problems are not limited to complete outages. Undervoltage, overvoltage, voltage fluctuations, and repeated power restoration can cause networking equipment to malfunction,

restart, or experience electrical stress. A suitable backup system should therefore provide not only power availability but also stable and regulated voltage. This is particularly important for sensitive electronic networking equipment.

2.7.6 NEED FOR ALTERNATIVE AND RENEWABLE BACKUP POWER

The limitations of unreliable grid electricity and conventional generators have increased interest in renewable-energy-based backup systems. Solar PV combined with battery storage can generate electricity during daylight and store energy for use when solar radiation is unavailable. Heinemann et al. (2022) identify standalone solar systems as a promising alternative to fossil-fuel generators in Nigeria, particularly because of their long-term economic and environmental benefits.

For small network installations, a DC solar generator can provide regulated DC power directly to compatible equipment instead of converting DC to AC and subsequently using individual AC adapters. This can reduce unnecessary conversion stages and improve the suitability of the system for low-power networking applications.

2.7.7 IMPORTANCE OF RELIABLE POWER FOR CLOUD AND NETWORK SERVICES

Modern organizations depend heavily on Internet connectivity, cloud computing, online applications, and remote communication. Network equipment provides the connection to these services; therefore, failure of a router, switch, access point, modem, or ONT can interrupt service even when the cloud platform itself remains operational. Uptime Institute (2024) identifies power-related failures as a major cause of IT service disruption.

Reliable backup power is therefore essential for maintaining network availability. The challenges discussed above support the proposed IoT-enabled solar generator with regulated 5 V, 9 V, and 12

V DC outputs, which combines solar generation, battery storage, DC voltage regulation, and IoT monitoring to provide a reliable alternative power source for selected networking devices.

2.8 INTERNET OF THINGS (IOT) TECHNOLOGY

The Internet of Things (IoT) refers to interconnected physical devices equipped with sensors, processing capabilities, communication interfaces, and software that enable them to collect, exchange, and process data over networks. NIST (2022) describes IoT systems as networked components that interact with physical entities through sensing and actuating capabilities. IoT therefore provides a link between physical systems and digital platforms for monitoring, analysis, and control.

2.8.1 COMPONENTS OF AN IOT SYSTEM

An IoT system generally consists of sensors, a processing or control unit, communication technology, and a monitoring or data-processing platform. Sensors collect information from the physical environment, controllers process the measurements, communication technologies transmit the data, and software platforms allow users to monitor and analyze the information (NIST, 2022).

2.8.2.1 SENSORS

Sensors provide the interface between the physical environment and the IoT system by measuring parameters such as temperature, voltage, current, pressure, and motion. In energy-management applications, sensors can monitor electrical variables and provide information about system performance (Amin et al., 2021).

For the proposed solar generator, voltage and current sensors are used to obtain information about the solar panel, battery, and connected load. These measurements provide the data required for real-time monitoring of the power system.

2.8.2.2 MICROCONTROLLERS AND IOT CONTROLLERS

Microcontrollers process sensor data and coordinate communication between sensors, electronic components, and monitoring platforms. They can perform calculations, identify operating conditions, and transmit information through a network. IoT controllers with integrated wireless communication are particularly suitable for small monitoring systems because they reduce the need for additional communication hardware (Hamdan et al., 2020).

Controllers such as the ESP8266 and ESP32 are suitable for the proposed system because they combine processing capability with Wi-Fi communication. The controller can receive voltage and current measurements, process the information, and transmit it to a remote monitoring platform.

2.8.2.3 COMMUNICATION TECHNOLOGIES

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IoT devices use communication technologies such as Wi-Fi, Bluetooth, Zigbee, cellular networks, LoRa/LoRaWAN, and Ethernet to exchange information. Selection depends on factors including range, bandwidth, power consumption, cost, and network availability (Saeed et al., 2023).

Wi-Fi is particularly suitable for the proposed system because it allows the IoT controller to connect to an Internet-accessible network and transmit solar-generator measurements to a remote platform. This provides the communication link between the power system and the user.

2.8.2.4 DATA TRANSMISSION AND MONITORING

Data transmission involves sending information collected by sensors from the IoT controller to a monitoring application, server, or cloud platform. The information can be stored, displayed, analyzed, or used to generate alerts. Local processing can also be performed before transmission to reduce communication requirements and response time (Hamdan et al., 2020).

For the proposed system, parameters such as solar-panel voltage, battery voltage, load current, and power conditions can be transmitted for remote monitoring. This allows users to observe the condition and availability of the solar generator without being physically present at the installation site.

2.8.3 IOT APPLICATIONS IN ENERGY MANAGEMENT

IoT technology is increasingly applied to energy management because it enables the measurement, communication, analysis, and control of energy-related information. Applications include monitoring renewable-energy generation, electricity consumption, battery conditions, equipment performance, and energy availability (Ahmad et al., 2021). Integration of IoT with renewable-energy systems can improve system visibility and support more effective energy management (Amin et al., 2021; Saeed et al., 2023).

In the proposed project, IoT provides the monitoring layer of the solar generator. Voltage and current sensors collect information from the PV panel, battery, and load, while an ESP8266/ESP32 controller processes the measurements and transmits them through Wi-Fi to an IoT or cloud platform. This enables remote monitoring of important system parameters and operating

conditions. The integration of IoT with solar generation, battery storage, and regulated DC power therefore provides a more intelligent approach to backup power for networking devices.

2.9 IOT-BASED MONITORING OF SOLAR GENERATOR SYSTEMS

IoT-based monitoring uses network-connected sensors, controllers, communication technologies, and software platforms to collect and transmit information about system conditions remotely. In solar generator systems, important parameters such as PV voltage, battery voltage, current, power, and battery condition can be monitored continuously. Monitoring these parameters supports early fault detection, performance assessment, and improved system reliability (Kalay, Kılıç, & Sağlam, 2022).

2.9.1 MONITORING OF BATTERY VOLTAGE

Battery voltage provides an indication of the charging and discharging condition of a solar generator. Continuous monitoring can identify low-voltage conditions, battery depletion, charging problems, or excessive loads. IoT-based PV systems have incorporated battery-voltage sensors to provide remote information about battery conditions (Mokom et al., 2025).

In the proposed system, the ESP32/ESP8266 can acquire battery voltage and use predefined thresholds to identify normal or low battery conditions.

2.9.2 MONITORING OF SOLAR PANEL VOLTAGE

Solar-panel voltage provides information about the availability and electrical condition of the PV source. Changes in irradiance, temperature, shading, and faults can affect PV output. Monitoring

PV voltage and current can therefore assist in identifying abnormal conditions and performance losses (Kalay et al., 2022).

For the proposed system, PV-voltage monitoring will provide information about solar-energy availability and the charging condition of the generator.

2.9.3 MONITORING OF CURRENT CONSUMPTION

Current monitoring determines the electrical current flowing to the load or between the PV system and battery. When combined with voltage measurement, current can be used to calculate power using:

$$P = V \times I$$

IoT-based PV monitoring studies have demonstrated the use of voltage and current measurements for real-time energy monitoring (Jabbar et al., 2022; Mokom et al., 2025). In the proposed system, load-current monitoring will help determine the electrical demand of connected routers, switches, modems, ONTs, and other networking devices.

2.9.4 MONITORING OF BATTERY STATE OF CHARGE

Battery State of Charge (SOC) represents the approximate amount of energy remaining in a battery. SOC can be estimated using parameters such as voltage, current, temperature, and charging or discharging history. Recent IoT-based battery-monitoring studies demonstrate the use of sensor measurements and computational methods to estimate battery condition.

For the proposed system, battery voltage and current measurements can provide the basic information required to estimate the battery charge condition and determine whether sufficient energy remains to operate the connected networking equipment.

2.9.5 REMOTE MONITORING OF POWER AVAILABILITY

Remote monitoring enables users to observe the condition of a solar generator without physically visiting the installation site. Parameters such as PV voltage, battery voltage, load current, power output, and battery status can be transmitted through Wi-Fi or other communication technologies to an Internet-based platform.

Jabbar et al. (2022) and Mokom et al. (2025) demonstrated remote monitoring of PV, battery, and load parameters using network-connected monitoring systems. This capability is relevant to the proposed project because users can remotely determine whether sufficient generated or stored energy is available for the networking devices.

2.9.6 REAL-TIME SYSTEM STATUS MONITORING WITH ALERTS AND NOTIFICATIONS

IoT monitoring can generate alerts when measured parameters exceed predefined limits. Examples include low battery voltage, excessive load current, insufficient PV voltage, and loss of power availability. Such notifications allow users to respond to developing problems before complete system failure occurs. Mokom et al. (2025) demonstrated the use of remote alerts for critical conditions in an IoT-based PV monitoring system.

In the proposed system, threshold-based alerts can therefore be used to notify users of low battery voltage, excessive current consumption, inadequate solar-panel output, or other abnormal conditions.

2.9.7 ADVANTAGES OF IOT-BASED MONITORING

IoT-based monitoring provides remote access, continuous data collection, real-time visualization, fault detection, historical data storage, performance analysis, and alerts. Cloud platforms can further allow users to access system information through web or mobile interfaces. These capabilities are particularly relevant to Network and Cloud Computing because the system combines renewable-energy generation with sensors, wireless communication, Internet connectivity, cloud-based data management, and remote access.

Overall, IoT provides the monitoring layer of the proposed solar generator. Voltage and current sensors collect information from the PV panel, battery, and load; an ESP32/ESP8266 controller processes the measurements and communicates them through Wi-Fi; and an IoT/cloud platform presents parameters such as battery voltage, PV voltage, current consumption, power availability, and battery status. Predefined thresholds can generate alerts when abnormal conditions occur. This integration supports remote management of the proposed 5 V, 9 V, and 12 V regulated DC solar generator and improves the reliability of power supplied to networking devices.

2.10 MICROCONTROLLERS IN IOT-BASED ENERGY SYSTEMS

Microcontrollers provide the processing and communication functions that connect sensors, electronic components, and IoT platforms. In energy systems, they acquire and process measurements such as voltage, current, power, and battery condition and transmit the information

for remote monitoring. Embedded controllers are therefore important in photovoltaic monitoring and data-acquisition systems (Kalay, Kılıç, & Sağlam, 2022).

2.10.1 ESP32 AND SIMILAR IOT PLATFORMS

The ESP8266 and ESP32 are widely used in IoT applications because they combine programmable processing with wireless communication. The ESP8266 provides a 32-bit processor, Wi-Fi, GPIO, ADC, SPI, I²C, UART, and PWM interfaces, while the ESP32 provides additional processing resources, Wi-Fi, Bluetooth/BLE, ADC, DAC, and multiple communication interfaces (Espressif Systems, 2025; 2026). These features make the ESP32 suitable for applications requiring sensor acquisition, data processing, and wireless communication using a single controller.

For the proposed solar generator, an ESP32/ESP8266 can receive voltage and current measurements, process the data, and transmit the results through Wi-Fi to an IoT or cloud platform.

2.10.2 SENSOR INTERFACING

Sensor interfacing enables a microcontroller to receive measurements from electrical and environmental sensors through interfaces such as ADC, GPIO, I²C, SPI, and UART. In PV monitoring systems, voltage and current measurements provide essential information about energy generation, battery operation, and load consumption (Kalay et al., 2022).

The proposed system uses voltage and current measurements to convert the electrical conditions of the solar panel, battery, and load into digital data that can be processed and monitored remotely.

2.10.3 VOLTAGE AND CURRENT MEASUREMENT

Voltage and current are fundamental measurements in an IoT-based energy system. Voltage sensors provide measurements within the microcontroller's acceptable input range, while current sensors measure the flow of electrical current through the PV, battery, or load circuits. The controller can calculate electrical power using $P = V \times I$.

PV monitoring research identifies voltage and current as important parameters for assessing system performance and detecting abnormal conditions (Kalay et al., 2022). IoT-based systems have also demonstrated the monitoring of PV, battery, and load voltage and current using microcontrollers and wireless communication (Mokom et al., 2025). In the proposed system, these measurements will provide information about solar generation, battery condition, and networking-device power consumption.

2.10.4 DATA PROCESSING AND CLOUD COMMUNICATION

After acquiring measurements, the microcontroller can convert raw sensor readings into meaningful values, calculate power, compare measurements with predefined limits, and prepare the data for transmission. Wi-Fi protocols such as MQTT or HTTP can then be used to send the information to a remote IoT or cloud platform. This makes the microcontroller the communication bridge between the physical solar generator and the remote monitoring environment.

2.10.5 AUTOMATIC MONITORING AND CONTROL

Microcontrollers can also support automatic monitoring and control by comparing measured parameters with predefined thresholds. For example, low battery voltage, excessive load current, or inadequate solar generation can trigger an alert or control action. ESP32-based energy-

monitoring systems have demonstrated the use of sensor measurements, cloud communication, and alerts for critical operating conditions (Mokom et al., 2025).

For the proposed IoT-enabled solar generator, the ESP32 therefore serves as the intelligent processing and communication unit connecting the solar power subsystem, regulated 5 V/9 V/12 V outputs, networking devices, Internet connection, and cloud-monitoring platform.

2.11 IOT COMMUNICATION AND CLOUD MONITORING

IoT communication and cloud monitoring provide the networking layer through which measurements from the solar generator can be transmitted, stored, visualized, and accessed remotely. The microcontroller can use Wi-Fi to connect to an Internet-accessible network and transmit parameters such as PV voltage, battery voltage, load current, power, and battery condition. Cloud connectivity allows users to access this information without being physically present at the installation.

2.11.1 MQTT AND OTHER IOT COMMUNICATION PROTOCOLS

Communication protocols define how IoT devices exchange information. MQTT is widely suited to IoT applications because its lightweight publish/subscribe architecture requires relatively low network resources. The OASIS MQTT Version 5.0 standard identifies MQTT as a lightweight protocol designed for machine-to-machine and IoT communication (OASIS, 2019).

In the proposed system, the ESP32 can publish solar-generator measurements to an MQTT broker while a monitoring dashboard subscribes to the relevant data. HTTP/HTTPS can also be used where direct web communication or API integration is required.

2.11.2 CLOUD-BASED MONITORING AND IOT DASHBOARDS

Cloud-based monitoring allows energy data to be stored, processed, and presented through Internet-accessible platforms. Dashboards can display parameters such as battery voltage, PV voltage, current, power, and system status using numerical indicators, charts, and historical records.

For the proposed project, cloud monitoring is important because it demonstrates the integration of energy monitoring, network communication, Internet connectivity, and cloud-based data management, which are central to Network and Cloud Computing.

2.11.3 REMOTE ACCESS, DATA VISUALIZATION, NOTIFICATIONS AND ALERTS

Remote access allows users to observe the solar generator from another location, while data visualization improves interpretation through graphs, numerical values, and status indicators. Monitoring platforms can also generate alerts when predefined conditions occur, such as low battery voltage, excessive current, or insufficient solar-panel output.

The proposed system can therefore combine real-time measurements, remote access, and notifications to provide early warning of conditions that could interrupt power to connected networking equipment.

2.11.4 IMPORTANCE OF CLOUD CONNECTIVITY FOR ENERGY MONITORING

Cloud connectivity extends energy monitoring from local measurement to remote data management. It allows measurements to be stored for historical analysis and accessed from multiple locations, supporting evaluation of solar generation, battery performance, and load

demand. This is particularly useful where networking equipment requires continuous and reliable power.

Consequently, cloud monitoring forms an important part of the proposed system by allowing users to remotely observe the condition of the 5 V, 9 V, and 12 V regulated DC solar generator and make informed decisions about available energy and connected network loads.

2.12 PROTECTION AND SAFETY OF SOLAR GENERATOR SYSTEMS

Protection and safety are important in solar generator systems because they combine PV generation, battery storage, DC-DC converters, and electrical loads. Protection is required against overvoltage, overcurrent, short circuits, reverse polarity, battery overcharging and over-discharge, and excessive temperature. Battery protection is particularly important because electrical and thermal abuse can reduce battery life and create safety risks (Chen et al., 2021).

2.12.1 OVERVOLTAGE, OVERCURRENT AND SHORT-CIRCUIT PROTECTION

Overvoltage protection prevents connected equipment from receiving voltage above its specified operating range, while overcurrent protection limits excessive current caused by overloads or faults. Short-circuit protection prevents potentially damaging currents from flowing through the battery, converters, wiring, and connected equipment. Fuses, circuit breakers, current-limiting circuits, and other protection devices can therefore be incorporated into the system. For the proposed generator, the 5 V, 9 V, and 12 V outputs should be appropriately regulated and protected according to their expected loads.

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2.12.2 REVERSE-POLARITY PROTECTION

Reverse-polarity protection prevents damage caused by incorrect connection of positive and negative terminals. Diodes, MOSFET-based circuits, and suitable connectors can be used to provide this protection. Its inclusion improves system safety during installation, maintenance, and connection of networking devices.

2.12.3 BATTERY OVERCHARGE AND OVER-DISCHARGE PROTECTION

Overcharging occurs when a battery receives energy beyond its safe charging limit, while over-discharge occurs when its voltage falls below the recommended minimum. Both conditions can reduce battery performance and service life and may create safety risks depending on the battery chemistry (Wang et al., 2023). A solar charge controller regulates charging and can disconnect the load when the battery reaches a low-voltage limit. These functions are essential because the proposed battery will supply networking devices when solar generation is insufficient.

2.12.4 THERMAL PROTECTION

Excessive temperature can accelerate battery degradation and increase safety risks, particularly in lithium-ion batteries. Thermal management therefore involves maintaining components within suitable operating temperature ranges through measures such as temperature monitoring, adequate ventilation, appropriate enclosure design, and automatic shutdown where necessary (Chen et al., 2021; Wang et al., 2021). These measures can improve the reliability and safety of the proposed solar generator.

2.12.5 PROTECTION OF CONNECTED NETWORKING DEVICES

Protection should also extend to the routers, switches, modems, ONTs, wireless access points, and other devices connected to the generator. Each device should receive its specified voltage, while the DC-DC converters should provide stable and appropriately protected outputs. Proper voltage regulation, fusing, current limitation, and short-circuit protection can reduce equipment damage and network downtime. Therefore, the proposed 5 V, 9 V, and 12 V DC outputs should be independently regulated and protected according to the expected load.

2.13 ENERGY MANAGEMENT FOR NETWORKING DEVICES

Energy management involves monitoring and efficiently utilizing available electrical energy to maintain essential loads while minimizing unnecessary consumption. In solar-powered backup systems, energy management is important because available energy depends on solar generation, battery capacity, weather conditions, conversion efficiency, and connected loads. Therefore, effective management requires balancing available energy with the demands of the connected networking devices.

2.13.1 ESTIMATING NETWORK-DEVICE ENERGY CONSUMPTION

Energy consumption can be estimated from the operating voltage, current, power, and duration of operation of each networking device. Power is expressed as ($P = VI$), while energy consumption depends on power and operating time. Since routers, switches, modems, ONTs, and wireless access points may have different power requirements, their combined load must be considered when selecting the solar panel, battery, and DC-DC converter ratings. The IEA (2025) reported that networking equipment such as switches, routers, and load balancers accounted for up to

approximately 5% of data-center electricity consumption in 2024, demonstrating the importance of understanding networking loads for effective power planning.

2.13.2 LOAD MANAGEMENT AND BATTERY MANAGEMENT

Load management involves adjusting connected loads according to available energy. When battery energy is limited, non-critical devices can be disconnected while essential networking devices remain powered. Battery management involves monitoring voltage, current, charging and discharging conditions to prevent excessive discharge and improve battery utilization. In the proposed system, the IoT controller can monitor battery and load conditions, generate low-battery alerts, and support the disconnection of selected non-critical loads. Prioritizing routers, modems, or ONTs over less critical devices can therefore extend backup duration and maintain essential network connectivity.

2.13.3 ENERGY CONSERVATION AND NETWORK AVAILABILITY

Energy conservation can be achieved through efficient networking equipment, appropriate power ratings, reduction of unnecessary loads, and continuous monitoring of energy consumption. IoT monitoring provides real-time information that can assist in identifying high-demand periods and improving energy utilization. The IEA (2023) also highlights the importance of efficiency improvements in IT infrastructure in limiting electricity-demand growth. For the proposed system, efficient energy management helps prevent premature battery depletion and improves the duration for which critical networking devices remain operational.

Overall, energy management complements IoT monitoring and electrical protection by ensuring that available solar-generated and stored energy is used effectively. Integrating these functions

provides a suitable foundation for an IoT-enabled solar generator with regulated 5 V, 9 V, and 12 V DC outputs for networking devices.

2.14 RELATED STUDIES ON IOT-BASED SOLAR AND BACKUP POWER SYSTEMS

Several studies have investigated solar energy, battery storage, IoT monitoring, cloud platforms, and energy management for renewable-energy and communication systems. However, most focus on PV monitoring, household energy management, or large telecommunications infrastructure rather than a compact solar backup system providing regulated 5 V, 9 V, and 12 V outputs for small networking devices.

2.14.1 JABBAR ET AL. (2022) – IOT-BASED SMART ENERGY METER FOR A STAND-ALONE PV SYSTEM

Jabbar et al. (2022) developed a LoRaWAN-based smart energy meter for a stand-alone PV system. The system used an Arduino, sensors, a LoRaWAN gateway, The Things Network, and the Althing Talk Maker cloud platform to monitor voltage, current, power, energy, and environmental parameters. The study demonstrated effective remote monitoring of PV-system parameters. However, it focused on energy measurement rather than dedicated backup power for networking equipment and used LoRaWAN rather than the Wi-Fi-based ESP32 architecture proposed in this project.

2.14.2 GAIKAR ET AL. (2023) – IOT-BASED SOLAR ENERGY MONITORING SYSTEM

Gaikar et al. (2023) investigated the application of IoT for remote monitoring and control of solar-energy systems. The study demonstrated how sensing, processing, communication, and Internet

technologies can improve access to solar-energy information. However, it did not specifically address networking-device backup power or regulated 5 V, 9 V, and 12 V DC outputs.

2.14.3 CONDON ET AL. (2023) – CLOUD-IOT-BASED ENERGY MANAGEMENT SYSTEM

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Condon et al. (2023) developed a Cloud-IoT-based Home Energy Management System using smart plugs, a smart meter, edge processing, and AWS IoT Core. The system collected and analyzed energy data and supported load control. Although designed for household energy management, the study demonstrates the relevance of IoT, cloud computing, data storage, and load management to the proposed solar-generator system.

2.14.4 JAVIDSHARIFI ET AL. (2021) – PV AND ENERGY STORAGE FOR GREEN BASE STATIONS

Javidsharifi et al. (2021) investigated PV and energy-storage sizing for cellular base stations using a multi-objective optimization approach. Their findings demonstrate the importance of correctly matching PV generation and battery capacity to network loads. However, the study focused on large cellular base stations and did not address IoT-based monitoring or low-voltage DC outputs for small networking devices.

2.14.5 SAHA ET AL. (2021) – BATTERY LIFETIME IN SOLAR-POWERED TELECOMMUNICATION NETWORKS

Saha et al. (2021) examined battery lifetime and PV-battery configurations for solar-powered telecommunications networks. The study showed that PV and battery sizing influence system reliability, cost, and outage probability. Its focus on cellular base stations and analytical modelling,

rather than small DC networking devices and IoT monitoring, limits its direct application to the proposed system.

2.14.6 BAIDAS ET AL. (2021) – SOLAR-POWERED CELLULAR BASE STATIONS

Baidas et al. (2021) examined the feasibility of solar-powered cellular base stations in Kuwait. The study demonstrated the potential of solar energy to reduce dependence on conventional electricity in telecommunications infrastructure. However, its focus was large cellular infrastructure rather than compact networking equipment, regulated DC outputs, or IoT/cloud monitoring.

2.14.7 ISLAM ET AL. (2023) – RENEWABLE-ENERGY-BASED OFF-GRID BASE STATIONS

Islam et al. (2023) proposed renewable-energy-based power architectures for off-grid heterogeneous cellular networks. Their approach combined solar PV, energy storage, energy sharing, and scheduling to improve energy utilization. Although relevant to renewable energy and network reliability, the study focused on cellular networks rather than the development of a compact DC solar generator for small networking devices.

2.14.8 MOKOM ET AL. (2025) – IOT MONITORING OF SOLAR PV SYSTEMS

Mokom et al. (2025) developed a low-cost IoT monitoring system for PV arrays, batteries, and loads using ESP32-based Internet connectivity. The system transmitted electrical measurements to an online platform and provided alerts for conditions such as low battery voltage and excessive load. This study is particularly relevant to the proposed project because it demonstrates the feasibility of combining solar generation, battery monitoring, ESP32 communication, cloud

monitoring, and alerts. However, it did not provide dedicated regulated 5 V, 9 V, and 12 V outputs for networking devices.

2.14.9 IOT-SCADA PV MONITORING USING ESP32 AND MQTT

Recent IoT-SCADA studies have demonstrated the use of ESP32, Wi-Fi, MQTT, Node-RED, cloud platforms, voltage and current sensors, and relay-based load control for PV monitoring. These architectures provide a close technical basis for the proposed communication pathway of sensors → ESP32 → Wi-Fi → cloud platform → remote monitoring and control. However, their main focus remains PV monitoring rather than the development of a dedicated regulated DC backup supply for networking equipment.

2.14.10 RENEWABLE ENERGY FOR TELECOMMUNICATIONS BASE STATIONS

Islam et al. (2021) reviewed renewable-energy-powered 5G infrastructure and identified renewable energy as a means of reducing grid dependence, operating costs, and environmental impact in cellular networks. The study considered renewable-powered base stations and energy-management strategies. However, it focused on large-scale cellular infrastructure rather than compact networking backup systems with multiple regulated DC outputs.

2.14.11 COMPARATIVE ANALYSIS AND RESEARCH GAP

The reviewed studies establish that solar energy, battery storage, IoT monitoring, wireless communication, cloud computing, and energy management can be effectively integrated into renewable-energy systems. Studies on cellular base stations demonstrate the importance of PV and battery sizing for reliable network operation, while IoT studies demonstrate the feasibility of remotely monitoring voltage, current, energy, battery conditions, and loads.

However, the reviewed systems generally focus on large telecommunications infrastructure, PV monitoring, or general energy management. They do not adequately combine a compact solar-battery backup generator, regulated 5 V, 9 V, and 12 V DC outputs, electrical protection, networking-device load management, and IoT/cloud-based remote monitoring in a single system. The proposed project therefore addresses this gap by developing an IoT-enabled solar generator specifically designed to provide reliable regulated DC backup power for routers, switches, modems, ONTs, wireless access points, and other small networking devices while enabling remote monitoring of important system parameters.

2.15 RESEARCH GAP

The reviewed literature demonstrates significant progress in integrating solar energy, battery storage, IoT monitoring, wireless communication, and cloud computing into renewable-energy systems. Existing studies have shown that parameters such as voltage, current, power, battery condition, and environmental variables can be monitored remotely using technologies such as ESP32, Wi-Fi, MQTT, and cloud platforms. However, most of these systems focus on photovoltaic monitoring, household energy management, or large telecommunications infrastructure rather than a compact backup power system specifically designed for small networking devices.

A major gap concerns the power requirements of routers, switches, modems, optical network terminals (ONTs), and wireless access points. Although renewable-energy research has addressed telecommunications infrastructure, much of it focuses on cellular base stations with substantially higher power requirements. There is therefore limited attention to compact solar-battery systems designed specifically for low-power networking equipment.

Another gap concerns the provision of multiple regulated DC outputs. Many conventional backup systems generate AC power, requiring separate adapters to convert the supply back to DC for networking equipment. This introduces additional conversion stages and may increase power losses. Existing PV monitoring systems also generally measure and report electrical parameters without providing simultaneously regulated 5 V, 9 V, and 12 V DC outputs for different networking devices.

The literature also shows a gap in the integration of energy monitoring, load management, and network continuity. Although some systems provide real-time monitoring and alerts, they are not specifically designed to determine available battery energy and support the continued operation of critical networking devices. A dedicated system could therefore prioritize essential loads and provide alerts when battery or load conditions approach critical levels.

The identified research gap can therefore be summarized as follows:

I. Application gap: Limited research focuses on solar backup systems specifically designed for small networking devices.
II. Power-output gap: Existing systems do not adequately provide simultaneous regulated 5 V, 9 V, and 12 V DC outputs for different networking loads.
III. Conversion gap: AC-based backup systems may introduce additional AC/DC conversion stages when supplying DC networking equipment.
IV. Monitoring gap: Some systems lack continuous remote monitoring of battery, load, and power conditions.
V. IoT/cloud gap: IoT-based PV monitoring is common, but fewer systems integrate cloud monitoring with a dedicated networking-device backup supply.

VI. Energy-management gap: Limited attention is given to prioritizing stored energy according to the importance of connected networking devices.

VII. Network-continuity gap: There is limited emphasis on maintaining routers, switches, modems, ONTs, and wireless access points during power interruptions using a purpose-built solar-battery DC system.

2.16.1 RESEARCH GAP ADDRESSED BY THE PROPOSED PROJECT

Based on these gaps, the proposed project develops an IoT-enabled solar generator specifically for networking devices. The system combines solar PV generation, battery storage, charge control, DC-DC voltage regulation, protection, and IoT/cloud monitoring. It is designed to provide regulated 5 V, 9 V, and 12 V DC outputs for compatible networking equipment while an ESP32/ESP8266-based controller monitors parameters such as solar-panel voltage, battery voltage, load current, and power conditions.

Through Wi-Fi and Internet connectivity, the measured information can be transmitted to an IoT/cloud platform for remote monitoring, visualization, and alerts. The proposed system therefore extends existing PV monitoring approaches by integrating power delivery, voltage regulation, networking-device support, energy monitoring, and cloud-based IoT functionality into a single compact backup system.

2.17 CONCEPTUAL FRAMEWORK OF THE PROPOSED SYSTEM

The conceptual framework illustrates the relationship between the major components of the proposed IoT-enabled solar generator for networking devices. It consists of a power generation and delivery pathway and an IoT monitoring and communication pathway.

The power pathway begins with the solar panel, which converts solar radiation into electrical energy. The energy is regulated by the solar charge controller and stored in the rechargeable battery. The stored energy is then supplied to DC-DC voltage regulators, which provide the required regulated 5 V, 9 V, and 12 V DC outputs for compatible networking devices such as routers, switches, modems, ONTs, and wireless access points.

The monitoring pathway uses voltage and current sensors to measure parameters such as solar-panel voltage, battery voltage, output voltage, and load current. An ESP32/ESP8266 IoT controller processes the measurements and transmits the information through Wi-Fi and the Internet to an IoT/cloud platform. The platform provides remote monitoring, data visualization, and alerts when predefined operating conditions occur.

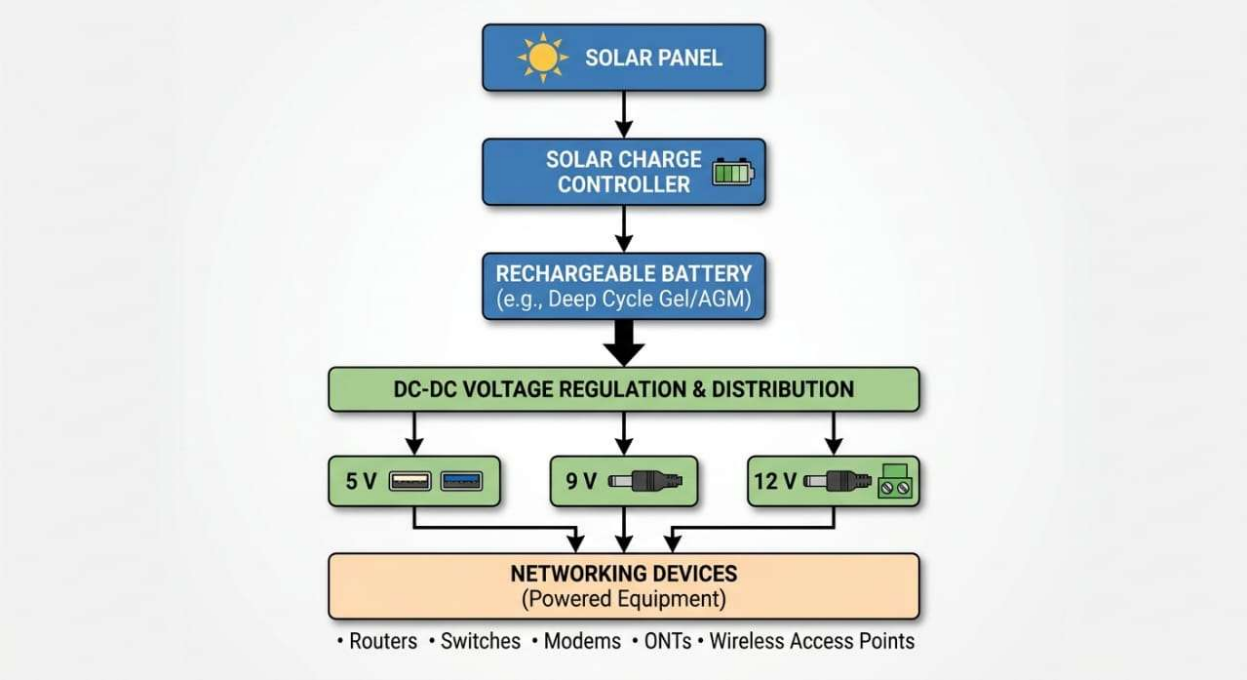


FIG. 2.1 CONCEPTUAL FLOW OF THE PROPOSED SYSTEM POWER GENERATION AND SUPPLY PATH

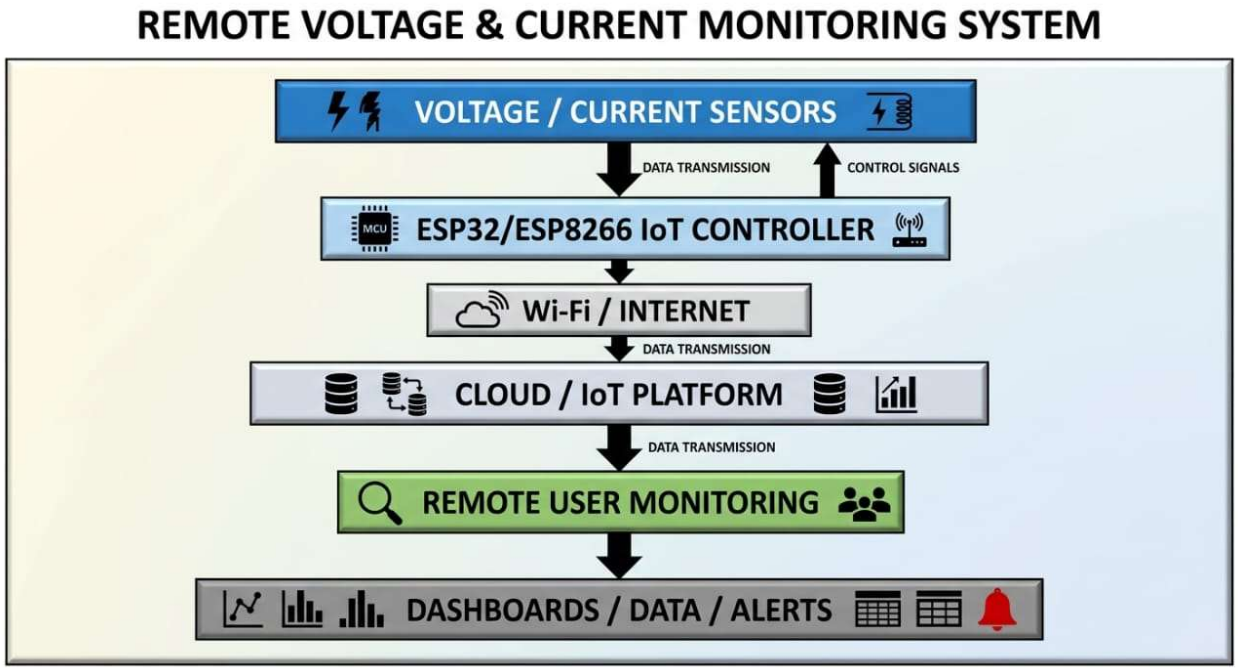


FIG. 2.2 CONCEPTUAL FLOW OF THE PROPOSED SYSTEM IOT MONITORING AND CLOUD PATH

2.18 SUMMARY OF LITERATURE REVIEW

The literature reviewed in this chapter established that solar generator systems can provide reliable alternative power by combining photovoltaic generation, battery storage, charge control, and power-conditioning technologies. Batteries provide energy storage, charge controllers regulate charging, and DC-DC converters provide suitable voltage levels for electronic loads. Research on DC energy systems also indicates the suitability of DC architectures for systems combining PV sources, batteries, and DC loads (Al-Ismail, 2021; Fan, Yu, & Xia, 2021).

The review further established that networking devices such as routers, switches, modems, ONTs, and wireless access points require stable and appropriately regulated DC power. Consequently, voltage regulation, overcurrent protection, short-circuit protection, reverse-polarity protection, and battery protection are important for reliable backup operation.

IoT technology provides an effective means of monitoring these systems. ESP32-based controllers can collect voltage and current measurements and transmit the information through Wi-Fi to cloud or IoT platforms. Recent PV monitoring studies using ESP32, MQTT, cloud platforms, and remote load control demonstrate the feasibility of real-time monitoring and control.

The review of related studies shows that previous researchers have successfully addressed individual aspects of the proposed system, including PV monitoring, battery management, renewable-energy telecommunications, IoT energy monitoring, cloud dashboards, and remote control. However, limited attention has been given to integrating these functions into a compact solar backup system specifically designed for small networking devices with multiple regulated DC outputs.

The proposed project therefore addresses this gap through the design and implementation of an IoT-enabled solar generator for networking devices with regulated 5 V, 9 V, and 12 V DC outputs. The system integrates solar generation, battery storage, charge control, DC-DC regulation, protection, networking-device power delivery, and IoT/cloud monitoring. This integration is intended to improve the availability and continuity of power for compatible networking devices while providing users with real-time information about important system operating conditions.

CHAPTER THREE

METHODOLOGY

The development of the IoT-Based Solar Generator for Networking Devices with Regulated 5 V, 9 V and 12 V DC Output was carried out using the System Development Life Cycle (SDLC). The SDLC provides a systematic approach to the development of a system through investigation, analysis, feasibility study, design, development, implementation and testing.

The stages adopted for the development of the system were:

I. Investigation/Preliminary Stage
II. System Analysis
III. Feasibility Study
IV. System Design
V. System Development
VI. System Implementation
VII. System Testing

Testing procedures and results are presented in Chapter Four.

3.1 INVESTIGATION/PRELIMINARY STAGE

This stage involved investigating the existing methods used to provide electrical power to networking devices such as routers, switches, modems and other low-voltage communication equipment.

The existing arrangement mainly depends on electricity supplied from the national grid. Networking devices are connected to the AC mains through their respective power adapters. During power outages, backup sources such as UPS units and fuel-powered generators may be used to maintain operation.

The investigation showed that frequent power interruptions can result in network downtime and interruption of communication services. Conventional backup systems may also require fuel, regular maintenance and manual supervision. In addition, conventional power arrangements do not normally provide integrated remote monitoring of parameters such as battery voltage, output voltage and load current.

Based on these observations, the need was identified for a solar-powered system capable of generating and storing electrical energy, providing regulated DC outputs and monitoring its operating parameters remotely.

The proposed system therefore combines a solar panel, charge controller, rechargeable battery, DC-DC voltage regulators, voltage and current sensors, ESP32 microcontroller and Blynk IoT platform to provide regulated 5 V, 9 V and 12 V DC power for compatible networking devices.

3.2 SYSTEM ANALYSIS

This stage involved analyzing the existing power supply arrangement, identifying its limitations and determining the requirements of the proposed system.

3.2.1 THE EXISTING SYSTEM

The existing system for powering networking devices depends primarily on the national grid. The AC supply is connected to networking equipment through individual AC-to-DC adapters.

When grid electricity is available, the networking devices operate normally. During power interruptions, the devices depend on available backup systems. UPS units provide stored electrical energy for a limited period, while fuel-powered generators can provide longer backup but require fuel and regular maintenance.

The existing arrangement also provides limited information about the condition of the power source and connected loads. Users may therefore need to physically inspect the equipment to determine the battery condition, output voltage or current consumption.

3.2.2 PROBLEMS OF THE EXISTING SYSTEM

The major problems identified in the existing system include:

I. Dependence on grid electricity, which may be interrupted.
II. Limited backup duration from conventional UPS systems.
III. Fuel consumption and maintenance requirements associated with generators.
IV. Lack of remote monitoring of electrical parameters.
V. Limited information about battery and load conditions.
VI. Manual supervision of backup power systems.
VII. Additional power conversion stages when AC mains is converted to DC for networking equipment.
VIII. Environmental effects associated with fuel-powered generators.

3.2.3 PROPOSED SYSTEM

The proposed system uses solar photovoltaic technology, battery storage, DC voltage regulation, an ESP32 microcontroller and IoT technology to provide a renewable and monitored power source for networking devices.

The system is designed to:

I. Generate electrical energy from solar radiation.
II. Store generated energy in a rechargeable battery.
III. Provide regulated 5 V, 9 V and 12 V DC outputs.
IV. Monitor battery voltage and load electrical parameters.
V. Transmit monitoring information through Wi-Fi.
VI. Display monitored parameters on the Blynk IoT platform.
VII. Provide suitable DC power to compatible networking devices.
VIII. Reduce dependence on grid electricity and fuel-powered generators.

3.3 FEASIBILITY STUDY STAGE

The feasibility study was carried out to determine whether the proposed system could be developed and operated using the available resources, components and technical skills.

3.3.1 OPERATIONAL COST

The proposed system is operationally feasible because it uses solar energy as its primary energy source and a rechargeable battery for energy storage. The stored energy is supplied to the DC voltage regulation section, which provides the required output voltages.

The ESP32 monitors the operating parameters and communicates with the Blynk IoT platform through Wi-Fi. This allows the operating condition of the system to be monitored without continuous physical inspection.

The system can therefore operate independently of grid electricity when sufficient stored solar energy is available.

3.3.2 ECONOMIC COST

The proposed system requires an initial investment in the solar panel, battery, charge controller, voltage regulators, ESP32, sensors and other components. However, the use of solar energy reduces dependence on fuel-powered generators and associated fuel costs.

The provision of regulated DC outputs also allows compatible networking devices to be supplied directly with DC power, reducing unnecessary conversion stages.

The system is therefore economically feasible for applications requiring backup power for low-voltage networking equipment.

3.3.3 TECHNICAL COST

The system is technically feasible because the required hardware and software components are available for development. The major hardware components include the solar panel, rechargeable battery, solar charge controller, LM2596 buck converters, ESP32 development board, voltage sensor and current sensor.

The software component uses the Arduino IDE and Blynk IoT platform. The ESP32 provides the processing and Wi-Fi communication required for remote monitoring.

The available components and technical skills required for circuit construction, microcontroller programming and IoT configuration were considered adequate for the development of the system.

3.4 SYSTEM DESIGN

The system was designed using a modular approach in which the complete system was divided into functional units. The major units are:

I. Solar energy generation unit.
II. Battery storage unit.
III. Solar charge-control unit.
IV. DC voltage regulation unit.
V. Voltage sensing unit.
VI. Current sensing unit.
VII. ESP32 control unit.
VIII. Wi-Fi communication unit.
IX. Blynk IoT monitoring unit.
X. DC output unit.

The system consists of two major sections: the power section and the monitoring section.

The power section consists of the solar panel, charge controller, rechargeable battery and LM2596 buck converters. The monitoring section consists of the voltage sensor, current sensor, ESP32 and Blynk IoT platform.

3.4.1 BLOCK DIAGRAM OF THE SYSTEM

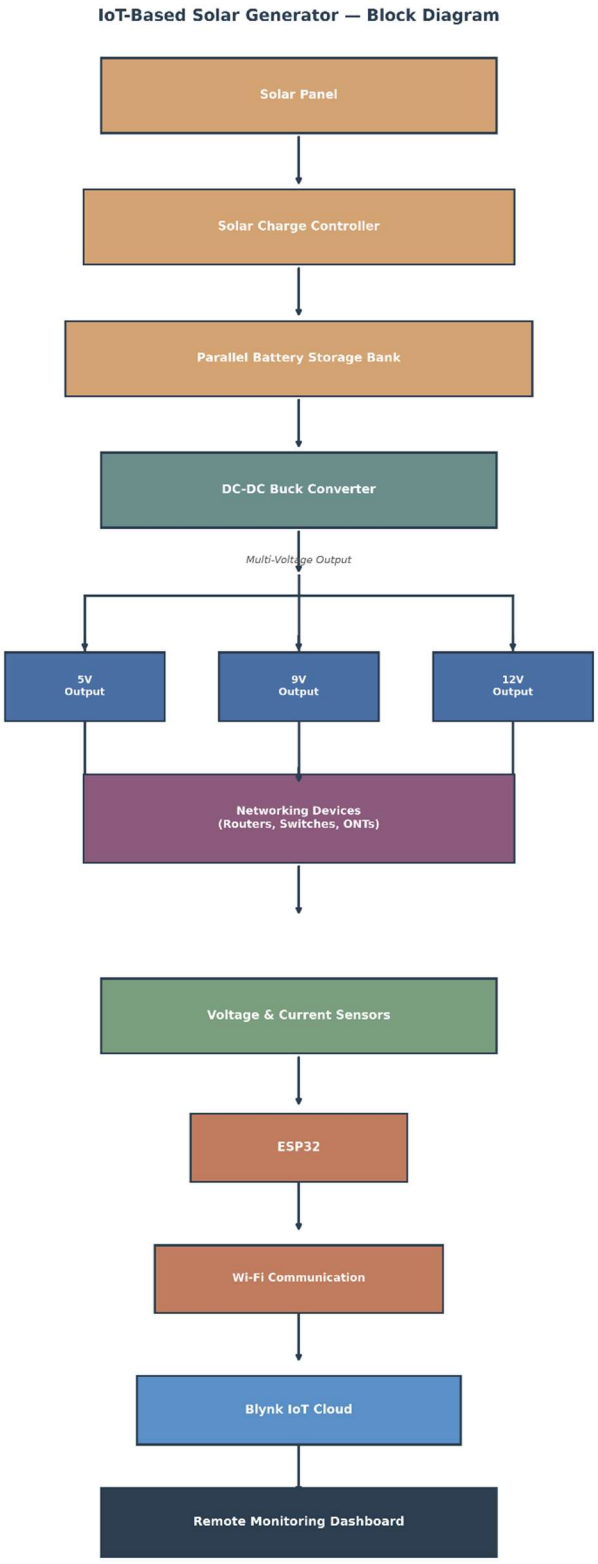


Fig. 3.1: Block Diagram of the Proposed IoT-Based Solar Generator

3.4.2 DESCRIPTION OF THE BLOCK DIAGRAM

The block diagram of the IoT-Based Solar Generator for Networking Devices with Regulated 5 V, 9 V and 12 V DC Output illustrate the flow of electrical energy from the solar source to the connected networking devices and the flow of monitoring data from the sensors to the IoT platform.

The solar panel serves as the primary source of electrical energy. It converts solar energy into DC electrical energy and supplies the generated power to the solar charge controller. The charge controller regulates the energy from the solar panel and controls the charging of the rechargeable battery. The battery stores the generated electrical energy for use when the solar panel is not producing sufficient power.

The stored DC energy from the battery is supplied to the DC regulation section, which consists of DC-DC buck converter modules. The regulation section converts the battery voltage into the required regulated output levels of 5 V, 9 V and 12 V DC. These regulated outputs are then supplied to compatible networking devices, such as routers, switches, modems and other low-voltage communication equipment.

In addition to the power path, the system has a separate monitoring path. A voltage sensor measures the voltage of the monitored power source, while a current sensor measures the current drawn by the connected load. The measurements from both sensors are supplied to the ESP32 microcontroller, which processes the sensor data.

The ESP32 uses its built-in Wi-Fi capability to transmit the processed information to the Blynk IoT platform through the Internet. The Blynk platform provides a remote monitoring interface

where the user can view the monitored electrical parameters and observe the operating condition of the solar generator.

Therefore, the block diagram consists of two interconnected paths. The power path provides the required electrical energy to the networking devices, while the monitoring path measures and communicates the operating parameters of the system. This arrangement enables the system to provide regulated DC power while also supporting remote monitoring through IoT technology.

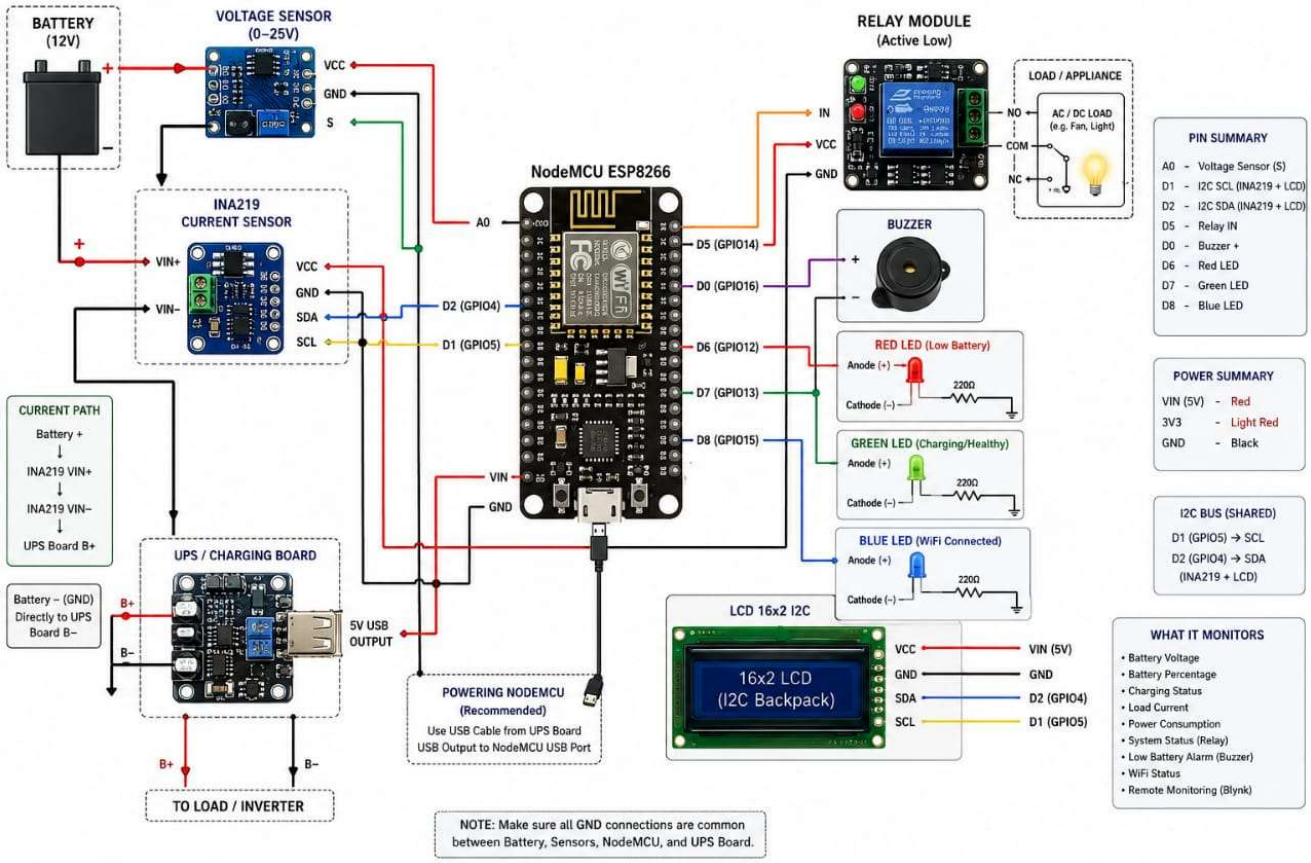


Fig. 3.2: Circuit Diagram of the Proposed IoT-Based Solar Generator

3.4.4 DESCRIPTION OF THE CIRCUIT DIAGRAM

The circuit diagram shows the electrical connections between the solar power generation, battery storage, voltage regulation, monitoring and output sections of the proposed system. The solar panel is connected to the input terminals of the solar charge controller. The charge controller regulates the electrical energy generated by the solar panel and supplies the controlled charging current to the rechargeable battery.

The rechargeable battery serves as the main energy storage and DC power source for the system. Its output is connected to the DC regulation section, which consists of LM2596 DC-DC buck converter modules. The buck converters regulate the battery voltage to the required levels for the connected networking devices. Three regulated DC outputs are provided, namely 5 V, 9 V and 12 V. Each output is connected to its appropriate DC output connector for supplying compatible networking equipment.

The monitoring section consists of a voltage sensor, INA219 current sensor and ESP32 microcontroller. The voltage sensor is connected to the voltage point being monitored and provides a scaled voltage signal to an analogue input of the ESP32. This enables the microcontroller to measure the corresponding voltage without directly exposing its input to the higher system voltage.

The INA219 current sensor is connected in the monitored load path to measure the current consumed by the connected networking equipment. The sensor communicates with the ESP32 through the I²C communication interface, allowing the measured current information to be processed by the microcontroller.

The ESP32 serves as the main monitoring and processing unit. It receives the voltage and current measurements from the sensors, processes the information and uses its built-in Wi-Fi capability to transmit the monitored data to the Blynk IoT platform. The Blynk platform provides the interface through which the user can remotely monitor the operating parameters of the solar generator.

LED indicators are incorporated to provide local visual indication of the system's operating status. The complete circuit and its associated components are installed inside a suitable protective enclosure, which provides physical protection for the electronic components, electrical connections and wiring.

Overall, the circuit is divided into a power section and a monitoring section. The power section is responsible for generating, storing, regulating and supplying DC power to the networking devices, while the monitoring section measures the electrical parameters and communicates the information to the Blynk IoT platform.

3.4.5 FLOWCHART OF THE OPERATION OF THE SYSTEM

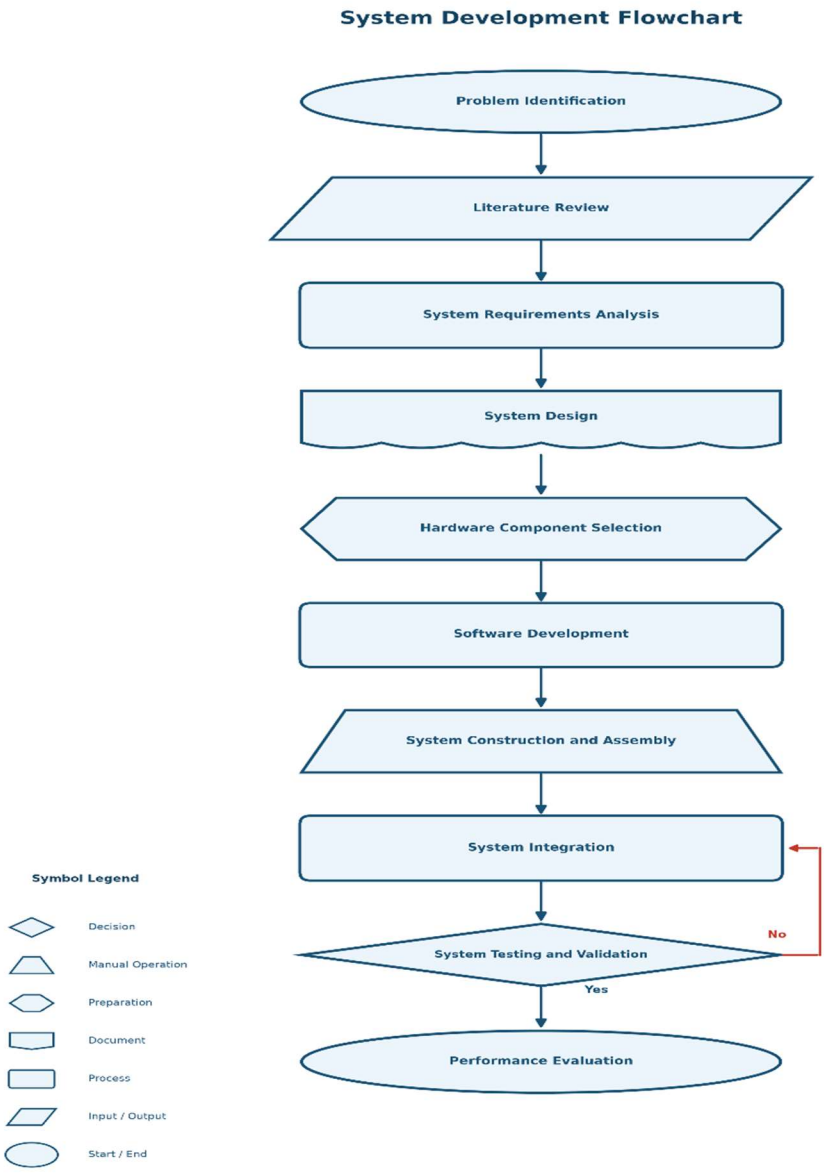


Fig. 3.3: Flowchart of the Operation of the Proposed System

3.4.6 DESCRIPTION OF THE FLOWCHART

The flowchart describes the sequence of operations of the proposed IoT-based solar generator from system start-up to continuous monitoring. When the system is switched on, the ESP32 microcontroller initializes the connected voltage and current sensors as well as the communication functions required for remote monitoring.

After initialization, the ESP32 attempts to establish a Wi-Fi connection. Once the connection is established, the controller connects to the Blynk IoT platform. The voltage and current sensors are then read at regular intervals to obtain the electrical parameters of the system.

The sensor readings are received and processed by the ESP32. The processed information is transmitted through the Wi-Fi connection to the Blynk IoT platform, where the monitored parameters are displayed on the configured IoT dashboard. This allows the operating condition of the solar generator and the connected load to be monitored remotely.

The sensing, processing and communication operations are performed continuously while the system remains powered. The ESP32 repeatedly obtains new sensor readings, processes the measurements and updates the information displayed on the Blynk dashboard.

At the same time, the power section operates to supply energy to the connected networking devices. Electrical energy generated by the solar panel passes through the solar charge controller and is stored in the rechargeable battery. The stored energy is then supplied to the DC regulation section, where it is converted into the required regulated 5 V, 9 V and 12 V DC outputs for the connected networking devices.

Thus, the flowchart represents two continuous operations: the power supply process, which provides regulated DC power to the networking devices, and the monitoring process, which measures system parameters and transmits the information to the Blynk IoT platform.

3.5 SYSTEM DEVELOPMENT

This phase involved the construction and programming of the proposed system. The hardware and software components were selected according to their functions, compatibility, availability and suitability for the project.

The following activities were carried out.

I. Circuit Design

The circuit was designed as separate power and monitoring sections.

The power section consists of the solar panel, solar charge controller, battery and LM2596 buck converters. The monitoring section consists of the voltage sensor, INA219 current sensor and ESP32.

Three LM2596 modules were used because the system requires three different regulated output voltages. Their adjustable output capability allows each module to be configured for 5 V, 9 V or 12 V.

II. Programming and Application Design

The ESP32 was programmed using the Arduino IDE and C/C++ programming language.

The program was developed to:

- Initialize the sensors.
- Measure voltage.
- Measure current.
- Process the sensor readings.
- Establish Wi-Fi communication.
- Connect to the Blynk IoT platform.
- Transmit monitoring data.
- Update the IoT dashboard.

Blynk was selected to provide the remote monitoring interface for the system.

III. Physical Construction

The components were arranged inside a suitable enclosure. The solar charge controller, battery, voltage regulators, ESP32, sensing modules and wiring were arranged to provide an organized physical structure.

The DC output connectors were positioned for easy connection of networking devices. The control and sensing components were arranged separately from the high-current power connections where necessary.

IV. Power Supply

The solar panel was used as the primary energy source. The solar charge controller was used to regulate the charging process and connect the solar panel to the rechargeable battery.

The battery supplies the stored DC energy to the voltage regulation section. The LM2596 modules regulate the battery voltage to the required 5 V, 9 V and 12 V outputs.

V. Control Unit

The ESP32 development board was selected as the control unit because it provides adequate processing capability and built-in Wi-Fi communication.

The ESP32 receives information from the voltage and current sensors, processes the measurements and transmits the information to the Blynk IoT platform.

An Arduino Uno was not used as the main controller because the ESP32 provides integrated Wi-Fi, which is required for the IoT monitoring function without requiring a separate wireless communication module.

VI. Sensor/Input Unit

A voltage sensor was used to measure the monitored voltage levels. The sensor provides a scaled signal that can be read by the ESP32.

An INA219 current sensor was used to measure load current. The INA219 was selected because it provides current measurement and communicates with the ESP32 using the I^2C interface.

VII. Display/Output Unit

The Blynk IoT dashboard provides the main remote display for the system. It displays the electrical parameters transmitted by the ESP32.

LED indicators provide local system-status indication. The regulated DC connectors provide the electrical output required by compatible networking devices.

VIII. Communication Unit

Wi-Fi communication is provided by the ESP32. No separate Wi-Fi module was required because Wi-Fi is integrated into the ESP32 development board.

The wireless connection allows the measured parameters to be transmitted to the Blynk IoT platform.

IX. Actuator Unit

The system does not require a conventional mechanical actuator. Its primary controlled outputs are the regulated DC power outputs.

The LM2596 buck converters regulate the battery voltage to the required 5 V, 9 V and 12 V levels. These outputs are supplied to the connected networking devices.

The LM2596 was selected because it is a switching DC-DC converter suitable for reducing DC voltage with greater efficiency than a conventional linear regulator.

3.6 IMPLEMENTATION

This stage involved assembling the hardware components, loading the developed software onto the ESP32 and integrating the power and monitoring sections into a complete system.

3.6.1 BUILDING AND DEVELOPMENT

The building and development of the proposed system involved the integration of the hardware and software components into a functional IoT-based solar generator. The construction was carried out systematically, beginning with the power generation and storage section before integrating the voltage regulation and monitoring sections.

The solar panel was connected to the input terminals of the solar charge controller. The charge controller was then connected to the rechargeable battery to regulate the charging process and store the electrical energy generated by the solar panel. The battery served as the main energy storage unit from which power was supplied to the rest of the system.

The output of the rechargeable battery was connected to the DC voltage regulation section. Three LM2596 DC-DC buck converter modules were incorporated into the system and adjusted to provide regulated 5 V, 9 V and 12 V DC outputs, respectively. These outputs were connected to the appropriate output terminals for supplying compatible networking devices.

The monitoring section was constructed by connecting the voltage sensor and current sensor to the power and load sections of the system. The sensors measure the relevant electrical parameters and send the measurement data to the ESP32 development board. The ESP32 processes the sensor data and prepares it for transmission.

The software component was developed using the Arduino IDE. The ESP32 was programmed to initialize the sensors, acquire electrical measurements, process the data, connect to the available Wi-Fi network and communicate with the Blynk IoT platform. The Blynk platform provides the interface through which the monitored parameters can be viewed remotely.

After the electrical connections and programming were completed, the components were arranged inside a protective enclosure. The wiring was organized and properly insulated to reduce the risk of short circuits and accidental contact between conductors. The arrangement also provided accessibility for maintenance and future inspection of the system.

The completed construction therefore integrated the solar energy generation, battery storage, DC voltage regulation, electrical parameter monitoring and IoT communication sections into a single system designed to provide regulated 5 V, 9 V and 12 V DC power to compatible networking devices while enabling remote monitoring through the Blynk IoT platform.

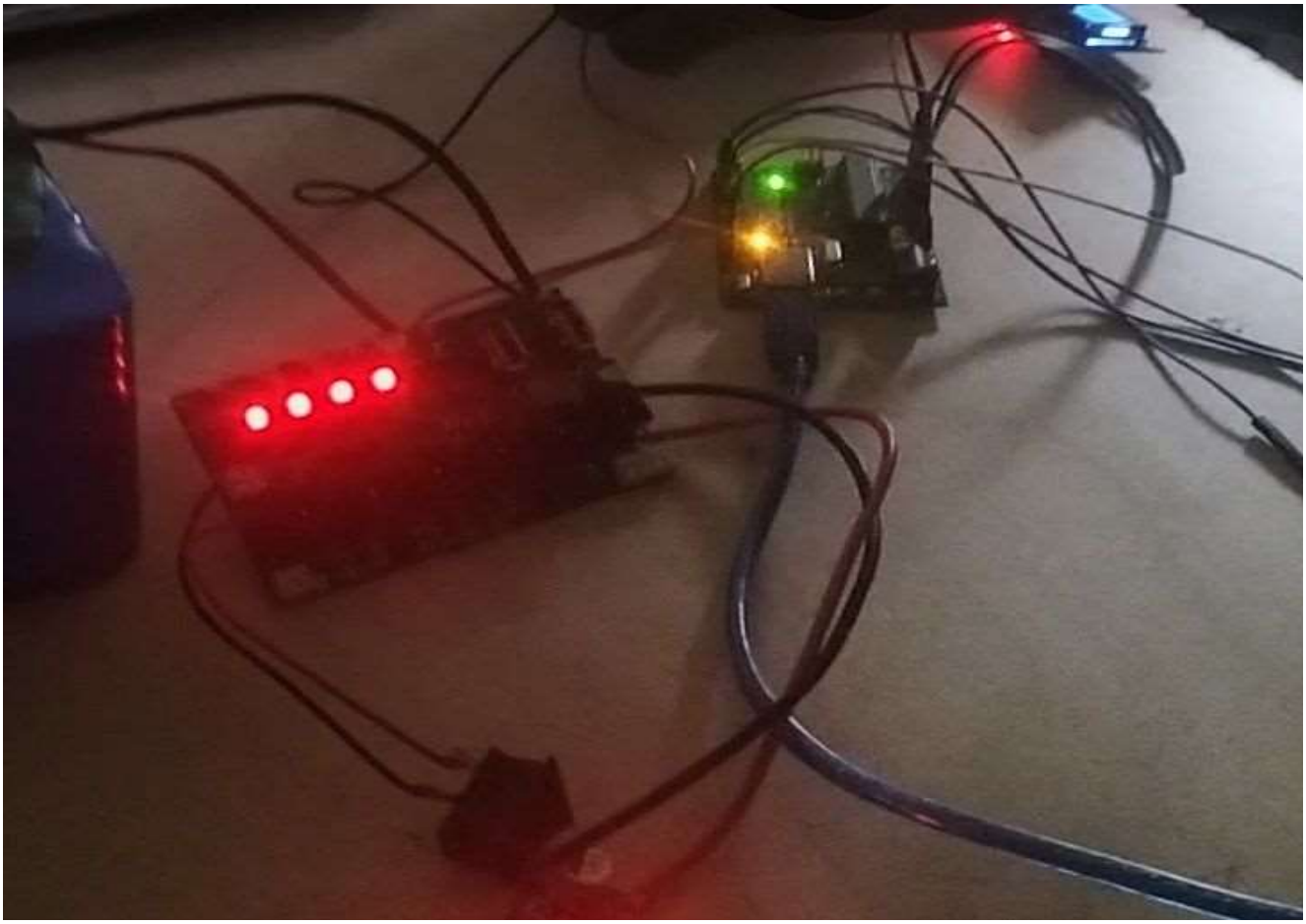


Fig. 3.4: Component arrangement during construction



Fig. 3.6: Installation of the ESP32 and sensing section

3.6.2 MATERIALS REQUIRED FOR THE COMPLETION OF THE TASK

3.6.2.1 SOFTWARE AND PROGRAMMING LANGUAGE USED

I. Arduino IDE

II. C/C++ Programming Language

III. Blynk IoT Platform

IV. ESP32 Board Support Package

V. Blynk and ESP32 Libraries

3.6.2.2 HARDWARE USED

I. Solar Panel

II. Rechargeable Battery

III. Solar Charge Controller

IV. LM2596 DC-DC Buck Converter for 5 V Output

V. LM2596 DC-DC Buck Converter for 9 V Output

VI. LM2596 DC-DC Buck Converter for 12 V Output

VII. ESP32 Development Board

VIII. DC Voltage Sensor

IX. INA219 Current Sensor

X. DC Output Connectors

XI. USB Output Ports

XII. LED Indicators

XIII. Connecting Wires and Heat-Shrink Tubing

XIV. Project Enclosure

3.7 SYSTEM CONSTRUCTION AND ASSEMBLY

The construction stage involved the physical integration of the selected components into a complete solar-powered networking system. The components were arranged inside a suitable enclosure to provide an organized structure, electrical protection, ease of maintenance, and proper heat management.

The solar panel was connected to the input terminals of the solar charge controller. The battery was connected to the battery terminals of the charge controller. The battery output was subsequently connected to the DC-DC buck converter section.

Each buck converter was individually adjusted to produce the required output voltage. One converter was configured for 5V, another for 9V, and the third for 12V.

The ESP32 was connected to the voltage and current sensing circuits. The sensing units provide electrical measurements to the ESP32, which processes the data before transmitting it through Wi-Fi to the Blynk IoT platform.

Proper electrical wiring, insulation, polarity, and component arrangement were considered during construction. Protection measures were incorporated to minimize the possibility of short circuits, excessive current, incorrect connections, and component damage.

The major construction procedures were:

I. COMPONENT ARRANGEMENT AND MOUNTING

All major components were arranged and mounted within the enclosure to produce a compact and organized system. Adequate spacing was provided around the power-conversion components to support heat dissipation and maintenance.

II. ELECTRICAL CONNECTION

The solar panel, charge controller, battery, DC-DC buck converters, voltage sensor, current sensor, ESP32, and output terminals were interconnected according to the designed system architecture.

III. Output Calibration

The three DC-DC buck converters were adjusted individually to obtain the required 5V, 9V, and 12V output voltages. A digital multimeter was used to verify the voltage levels before connecting networking devices.

IV. ESP32 Programming and Configuration

The ESP32 was programmed using the Arduino Integrated Development Environment (Arduino IDE). The firmware was configured to read the connected sensors, process electrical measurements, establish Wi-Fi communication, and transmit data to the Blynk IoT platform.

3.7.1 Blynk IoT Operation

The Blynk IoT platform provides the remote monitoring interface for the developed system. The ESP32 communicates with the Blynk cloud platform through Wi-Fi and sends measured electrical parameters to the configured dashboard.

The dashboard displays important system parameters such as:

I. Battery voltage.

II. Output voltage.

III. Load current.

IV. Power consumption.

V. System operating status.

11 The integration of Blynk eliminates the need for continuous physical inspection of the system and provides a convenient method of monitoring system performance remotely.

CHAPTER FOUR

SYSTEM IMPLEMENTATION, OPERATION, RESULTS, AND TESTING

4.1 DESCRIPTION OF THE SYSTEM

This chapter presents the implementation, operation, input design, output design, results, testing, and evaluation of the developed IoT-Based Solar Generator for Networking Devices with Regulated 5V, 9V, and 12V DC Outputs. The chapter describes how the hardware and software components were integrated to produce a functional renewable-energy power supply system for low-voltage networking equipment.

The developed system is designed to provide a reliable alternative power source for networking devices such as routers, switches, optical network terminals (ONTs), modems, and other communication equipment that operate with low-voltage DC power. The system combines solar energy generation, battery storage, DC voltage regulation, and Internet of Things (IoT) technology to provide both power delivery and remote monitoring.

The power section of the system consists of a photovoltaic (PV) solar panel, solar charge controller, rechargeable battery storage, and DC-DC buck converter modules. The solar panel converts solar radiation into electrical energy, which is regulated by the solar charge controller before being stored in the battery. The stored energy is subsequently supplied to the DC-DC buck converters, which regulate the battery voltage to the required 5V, 9V, and 12V DC output levels.

The monitoring section is built around an ESP32 development board. Voltage and current sensors are connected to the ESP32 to measure important electrical parameters of the system. The ESP32 processes the sensor readings and transmits the information through Wi-Fi to the Blynk IoT

platform. The Blynk dashboard provides a means of remotely observing the operating condition of the solar generator.

The major components of the developed system include:

I. Photovoltaic solar panel.
II. Solar charge controller.
III. Rechargeable battery bank.
IV. DC-DC buck converter modules.
V. ESP32 development board.
VI. Voltage sensor.
VII. Current sensor.
VIII. 5V DC output terminal.
IX. 9V DC output terminal.
X. 12V DC output terminal.
XI. Protection and wiring components.
XII. Networking-device loads.

The system is designed so that the energy flow and monitoring functions operate together. The power subsystem provides the required electrical energy, while the ESP32-based monitoring subsystem measures and communicates the condition of the system.

The integration of these components provides an intelligent DC power system capable of supplying regulated power to networking devices while allowing the user to remotely observe parameters such as battery voltage, output voltage, load current, power consumption, and system operating status.

System Architecture

The overall architecture of the developed system can be represented as follows:

Solar Panel → Solar Charge Controller → Battery Bank → DC-DC Buck Converters → 5V / 9V / 12V DC Outputs → Networking Devices

The monitoring path operates simultaneously as:

Voltage and Current Sensors → ESP32 → Wi-Fi → Blynk IoT Dashboard

The power and monitoring subsystems therefore operate together. The power subsystem is responsible for energy generation, storage, regulation, and distribution, while the monitoring subsystem is responsible for measurement, processing, communication, and remote visualization.

4.2 OPERATIONAL DESCRIPTION

The developed solar generator is operated by first exposing the photovoltaic panel to available sunlight. The solar panel converts solar radiation into DC electrical energy and supplies the generated energy to the solar charge controller.

The solar charge controller regulates the electrical energy received from the solar panel and controls the charging of the rechargeable battery. This ensures that the generated energy is properly transferred to the battery for storage.

When power is required by a networking device, energy stored in the battery is supplied to the DC-DC buck converter section. Three buck converter modules are configured to provide regulated

5V, 9V, and 12V DC outputs. The appropriate output terminal is selected according to the voltage requirement of the networking device.

A networking device is connected only to the output voltage specified by its manufacturer. For example, a compatible 5V device is connected to the 5V output, while equipment requiring 9V or 12V is connected to the corresponding output terminal.

During operation, the voltage and current sensors continuously measure electrical parameters within the system. The ESP32 receives these measurements, processes the sensor readings, and communicates with the Blynk IoT platform through Wi-Fi.

The Blynk dashboard displays the collected information, allowing the user to monitor the solar generator without physically inspecting the system. The monitored parameters include battery voltage, output voltage, load current, power consumption, and general system status.

The general operational sequence is as follows:

I. The solar panel receives solar radiation.
II. The solar panel converts solar energy into electrical energy.
III. The solar charge controller regulates the generated energy.
IV. The regulated energy charges the rechargeable battery.
V. Stored battery energy is supplied to the DC-DC buck converters.
VI. The buck converters regulate the battery voltage to 5V, 9V, and 12V.
VII. The appropriate output is connected to a compatible networking device.
VIII. Voltage and current sensors measure the electrical parameters.
IX. The ESP32 receives and processes the sensor measurements.

X. The ESP32 establishes Wi-Fi communication.

XI. The measured information is transmitted to the Blynk IoT platform.

XII. The user observes the system parameters through the Blynk dashboard.

XIII. The sensing, processing, power supply, and communication operations continue while the system is operating.

4.3 SYSTEM CODE

The system code was developed using the Arduino Integrated Development Environment (Arduino IDE). The program controls the sensing, processing, Wi-Fi communication, and Blynk IoT monitoring functions of the ESP32-based system.

The program performs the following major operations:

I. Initializes the ESP32 and connected sensors.

II. Establishes communication with the voltage and current sensing units.

III. Reads electrical measurements at defined intervals.

IV. Processes the measured voltage and current values.

V. Establishes a Wi-Fi connection.

VI. Connects the ESP32 to the Blynk IoT platform.

VII. Transmits measured data to the Blynk dashboard.

VIII. Continuously updates the displayed system parameters.

The complete source code developed for the ESP32-based monitoring system is provided in **Appendix B**.

4.4 INPUT DESIGN

The input design describes the sources of energy and information entering the developed system. The major energy input is solar energy received by the photovoltaic panel, while the monitoring inputs are obtained from the voltage and current sensors.

4.4.1 Solar Energy Input

The photovoltaic panel serves as the primary energy input to the system. It receives solar radiation and converts it into electrical energy in the form of DC voltage and current.

The generated electrical energy is supplied to the solar charge controller. The charge controller regulates the energy before transferring it to the battery storage system.

4.4.2 Battery Input

The rechargeable battery serves as the energy storage source for the DC regulation section. Energy stored during periods of solar availability is supplied to the buck converters when networking devices require power.

The battery therefore enables the system to continue supplying power when solar radiation is insufficient or unavailable.

4.4.3 Voltage Sensor Input

The voltage sensor provides the ESP32 with information about the voltage being monitored. The measured voltage is processed by the ESP32 and used to determine the electrical operating condition of the system.

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4.4.4 Current Sensor Input

The current sensor provides information about the amount of electrical current consumed by the connected load. The ESP32 receives the current measurement and processes the information before transmitting it to the Blynk IoT platform.

4.5 OUTPUT DESIGN

The output design describes the regulated electrical outputs provided by the system and the information presented through the IoT monitoring platform.

4.5.1 Regulated 5V DC Output

The 5V output is provided through the designated 5V USB output terminal. It is intended for compatible networking and communication equipment that requires a regulated 5V DC input.

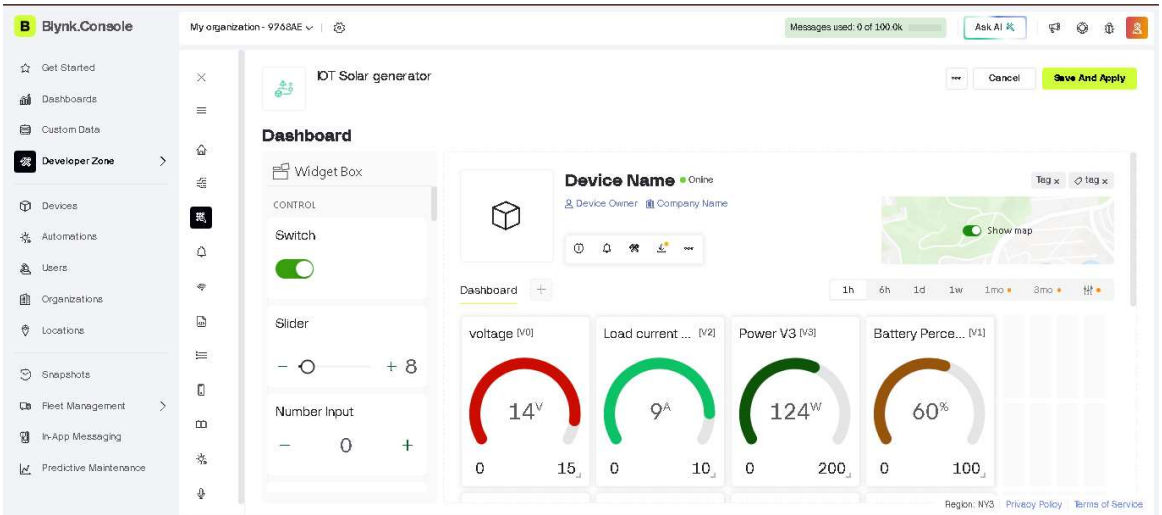


Figure 4.1: Blynk Development Environment

4.5.2 Regulated 9V DC Output

The 9V output is provided through the designated 9V DC terminal. It is intended for compatible routers, modems, and other networking devices that require 9V DC input.

4.5.3 Regulated 12V DC Output

The 12V output is provided through the designated 12V DC terminal. It is intended for compatible networking equipment such as routers, switches, ONTs, and other devices requiring 12V DC input.

The voltage rating of the networking device must always be checked before connection to prevent equipment damage.

4.5.4 IOT MONITORING OUTPUT

In addition to the electrical power outputs, the system provides digital monitoring information through the Blynk IoT dashboard.

The monitoring interface provides information such as:

I. Battery voltage.
II. Output voltage.
III. Load current.
IV. Power consumption.
V. System operating status.

The monitoring output allows the user to observe the operating condition of the solar generator remotely through an Internet-connected device.

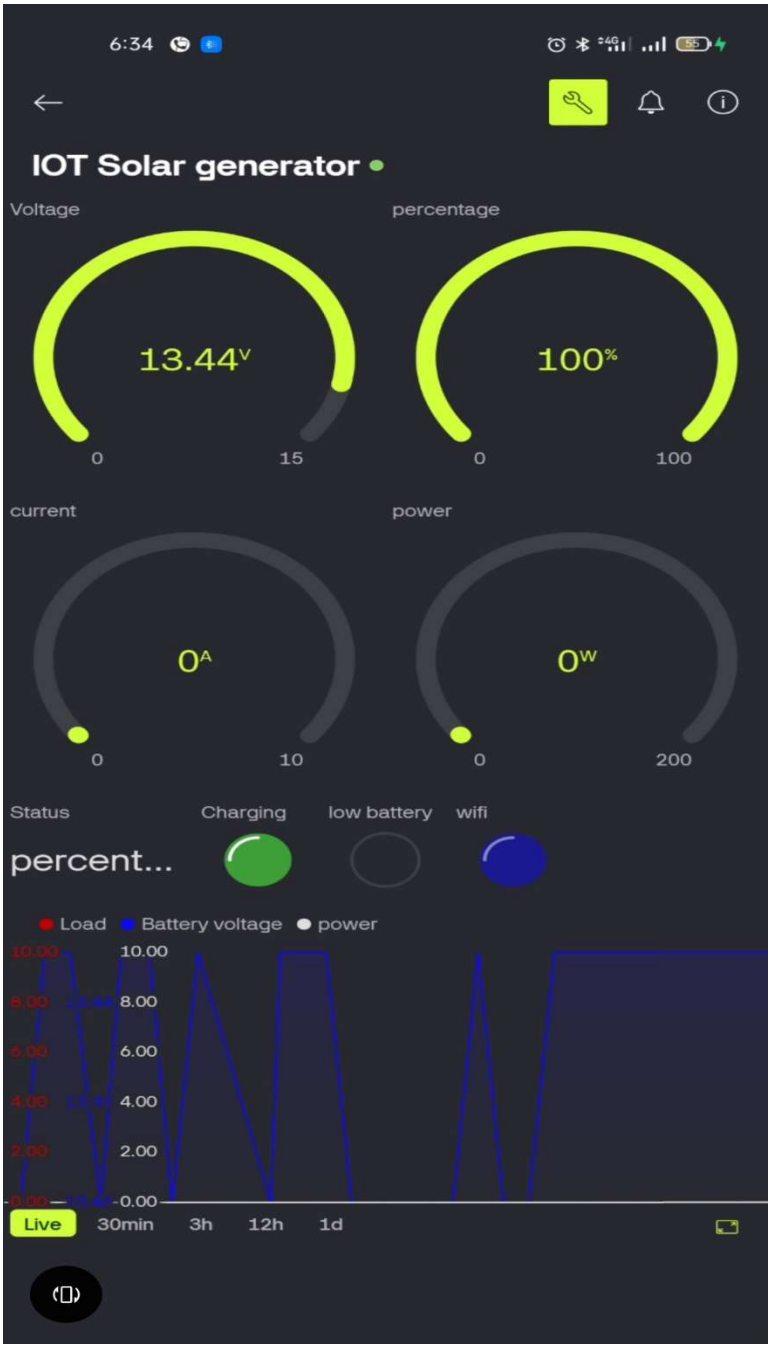


Fig 4.1: IoT Monitoring Output

4.6 SYSTEM EVALUATION AND TESTING

System evaluation and testing were carried out to determine the functionality, stability, reliability, and effectiveness of the developed IoT-based solar generator. The testing process involved evaluating the individual subsystems as well as the complete system under different operating conditions.

The major tests carried out were solar charging, battery performance, DC output voltage regulation, networking-device load operation, and IoT monitoring.

4.6.1 Solar Charging Test

The solar charging test was performed to determine whether the photovoltaic panel and solar charge controller could successfully generate, regulate, and transfer electrical energy to the battery.

The solar panel was exposed to available sunlight and connected to the solar charge controller. The voltage generated by the panel and the charging behavior of the battery were monitored during the test.

The successful transfer of energy from the solar panel through the charge controller to the battery confirmed the operation of the solar charging subsystem.

4.6.2 Battery Performance Test

The battery performance test evaluated the ability of the rechargeable battery to store and supply electrical energy. The battery voltage was monitored during charging and while supplying power to connected loads.

The test confirmed that stored energy could be supplied to the DC-DC converter section when solar energy was unavailable. This demonstrates the ability of the battery to serve as the energy storage component of the system.

4.6.3 DC Output Voltage Regulation Test

The three LM2596 step-down buck converter modules were tested to determine their ability to provide stable 5V, 9V, and 12V DC outputs.

Each converter was individually calibrated using a digital multimeter. The onboard potentiometer was adjusted until the required output voltage was obtained.

89 The calibration results are presented in Table 4.1.

Table 4.1: DC Output Voltage Calibration Results

The results show that the three output channels produced voltages very close to their respective target values. The highest observed deviation was 0.4%, which is within the specified ±2% tolerance.

The calibrated outputs were subsequently subjected to load testing to determine their voltage stability under different current conditions.

The results indicate that the three DC output channels maintained relatively stable voltage levels under the tested load conditions. The 5V output decreased from 4.98V at 0.5A to 4.96V at 1.0A. Similarly, the 9V output decreased from 8.95V at 0.3A to 8.92V at 0.7A, while the 12V output remained highly stable at 12.03V and 12.00V under the tested loads.

The voltage deviations remained within the specified tolerance. The results therefore demonstrate that the LM2596 buck converters were capable of providing suitable regulated DC outputs for compatible networking equipment under the tested conditions.

The measured efficiency values ranged from 91.8% to 94.2%, indicating that the converters provided relatively efficient voltage conversion from the battery supply.

4.6.4 Networking Device Load Test

A load test was conducted using compatible networking equipment and representative electrical loads. The purpose of the test was to determine whether the developed system could supply stable DC power while networking devices were operating.

The appropriate output voltage was selected according to the rated input requirement of each device. During the test, output voltage, load current, and operating behavior were observed.

The test demonstrated that the developed system could supply regulated DC power through the 5V, 9V, and 12V output channels to compatible networking devices.

The results also emphasized the importance of matching the output voltage of the solar generator with the specified input voltage of each networking device.

4.6.5 IoT Monitoring Test

The IoT monitoring test was conducted to verify communication between the ESP32 microcontroller and the Blynk IoT platform.

The system was tested to confirm that the ESP32 could:

I. Establish a Wi-Fi connection.

II. Connect to the Blynk IoT platform.

III. Read voltage and current sensor values.

IV. Transmit the data to the Blynk dashboard.

V. Display the parameters remotely.

The successful transmission and visualization of the sensor readings confirmed the operation of the IoT monitoring subsystem.

4.6.6 Real-Time Data Transmission and Visualization

The IoT monitoring functionality was evaluated by observing the transmission of system data from the ESP32 to the Blynk cloud platform during system operation.

The ESP32 periodically collected sensor readings and transmitted the processed information through the available Wi-Fi network. The Blynk dashboard displayed the measured parameters, allowing the user to observe the operating condition of the solar generator remotely.

The successful display of the measured parameters confirmed that the monitoring subsystem could provide remote information about the system's electrical condition.

4.6.7 Battery State-of-Charge Estimation

The battery state of charge (SoC) was estimated using battery terminal voltage. A voltage-to-SoC relationship was used to provide an approximate indication of the remaining battery capacity.

Output	Load Current (A)	Output Voltage (V)	Voltage Ripple (mV)	Efficiency (%)
5V USB	0.5	4.98	45	92.3
5V USB	1.0	4.96	58	91.8
9V DC	0.3	8.95	52	93.1
9V DC	0.7	8.92	67	92.5
12V DC	0.5	12.03	38	94.2
12V DC	1.0	12.00	51	93.8

Table 4.1: DC Output Performance Under Load Conditions

Battery Voltage (V)	Estimated State of Charge (%)	System Status
≥ 12.6	100	Fully Charged
12.4	90	Good
12.2	75	Good
12.0	50	Fair
11.8	25	Low – Warning
≤ 11.5	10	Critical – Alarm

Table 4.2: Voltage-to-SoC Mapping

The voltage-based estimation provides the user with an approximate indication of battery condition through the monitoring interface. Since battery terminal voltage can vary depending on load, charging condition, temperature, and battery chemistry, the values should be regarded as an indication rather than an exact measurement of the remaining battery capacity.

4.6.8 Overall System Test Summary

The major test results obtained from the developed system are summarized in Table 4.3.

Test Parameter	Expected Result	Observed Result	Status
Solar charging	Battery receives regulated charging energy	Charging operation observed	Pass
Battery storage	Battery stores and supplies energy	Battery supplied the regulation section	Pass
5V output	Approximately 5V	5.02V calibrated	Pass
9V output	Approximately 9V	8.97V calibrated	Pass
12V output	Approximately 12V	12.05V calibrated	Pass
Output load regulation	Stable output under load	Stable output observed	Pass
ESP32 sensing	Sensor values received	Measurements processed	Pass
Wi-Fi communication	ESP32 connects to network	Communication established	Pass
Blynk monitoring	Data displayed remotely	Parameters displayed	Pass
Networking load	Compatible devices receive power	Regulated power supplied	Pass

Table 4.4: Overall System Test Summary

The results indicate that the major subsystems of the developed solar generator operated according to their intended functions.

4.6.9 Achievement of Research Objectives

The testing and evaluation results demonstrate that the major objectives of the project were achieved.

Objective 1: To integrate a solar-powered battery charging subsystem with regulated DC outputs suitable for networking devices.

Achievement: The system successfully integrated a photovoltaic panel, solar charge controller, rechargeable battery, and three regulated DC output channels providing 5V, 9V, and 12V.

Objective 2: To implement a multi-output DC power distribution system capable of supplying networking devices.

Achievement: The system provided regulated 5V, 9V, and 12V DC outputs. The measured output voltages remained close to their specified target values during calibration and load testing.

Objective 3: To develop an IoT-based real-time monitoring system for remote performance assessment.

Achievement: An ESP32-based monitoring subsystem was successfully integrated with the Blynk IoT platform. The system enabled remote observation of electrical parameters, including voltage and current.

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4.6.10 System Limitations

Although the developed system achieved its intended functions, certain limitations were identified during implementation and testing.

First, the duration for which the system can supply networking devices without solar input depends on the capacity and condition of the battery. Extended operation during periods of low solar radiation may therefore reduce the available backup time.

Second, the remote IoT monitoring function depends on the availability of a suitable Wi-Fi network and Internet connection. If the network connection is unavailable, the system can continue to supply DC power, but remote monitoring through the Blynk platform may not be available.

Third, the battery state-of-charge estimation based primarily on terminal voltage provides an approximate indication rather than a highly precise measurement of actual remaining battery capacity.

Finally, the available power capacity limits the number and type of networking devices that can be connected simultaneously. Users must ensure that the total load does not exceed the capacity of the battery, converters, wiring, and protection components.

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 SUMMARY

This project focused on the design and implementation of an IoT-based solar generator for networking devices with regulated 5V, 9V, and 12V DC outputs. The project was developed in response to the challenges associated with unreliable electricity supply and the need for continuous power availability for networking equipment such as routers, switches, modems, optical network terminals (ONTs), and other low-voltage communication devices.

The developed system combines solar energy generation, battery storage, voltage regulation, and Internet of Things (IoT) technology into a single power supply system. The major hardware components include a photovoltaic solar panel, solar charge controller, rechargeable battery, DC-DC buck converters, ESP32 development board, voltage sensor, current sensor, and regulated output terminals.

The solar panel serves as the primary energy source by converting solar radiation into electrical energy. The solar charge controller regulates the energy from the panel and controls the charging of the rechargeable battery. The stored energy is then supplied to the DC-DC buck converters, which regulate the battery voltage to the required 5V, 9V, and 12V DC output levels.

An ESP32 microcontroller was incorporated into the system to provide monitoring and communication functions. The voltage and current sensors provide electrical measurements to the ESP32, which processes the information and transmits the measured parameters through Wi-Fi to the Blynk IoT platform. This enables the user to remotely monitor parameters such as battery voltage, output voltage, load current, power consumption, and system operating status.

The developed system was subjected to various tests, including solar charging, battery performance, DC output voltage regulation, networking-device load operation, IoT monitoring, remote monitoring, and battery state-of-charge estimation. The test results showed that the system successfully performed its intended functions. The regulated outputs remained close to their required voltage levels under the tested load conditions, while the IoT monitoring subsystem successfully transmitted and displayed system parameters through the Blynk dashboard.

Overall, the project demonstrates the integration of renewable energy, DC voltage regulation, and IoT monitoring to provide a functional backup power solution for compatible networking devices.

5.2 CONCLUSION

The design and implementation of the IoT-based solar generator for networking devices with regulated 5V, 9V, and 12V DC outputs was successfully achieved. The developed system provides a renewable-energy-based alternative for supplying power to compatible networking devices during periods of unreliable or unavailable grid electricity.

The integration of the photovoltaic panel, charge controller, battery storage, and DC-DC buck converters enabled the system to generate, store, and regulate electrical energy into the required DC voltage levels. The testing results confirmed that the system was capable of supplying regulated 5V, 9V, and 12V outputs to compatible loads with satisfactory performance.

The incorporation of the ESP32 microcontroller, voltage and current sensors, Wi-Fi communication, and Blynk IoT platform further improved the functionality of the system by providing remote monitoring of important electrical parameters. The successful transmission and display of the measured parameters confirmed that the monitoring subsystem was able to provide useful information about the operating condition of the solar generator.

The project therefore demonstrates that combining solar energy generation, battery storage, regulated DC power conversion, and IoT-based monitoring can provide an effective and intelligent backup power solution for low-voltage networking equipment. The developed system can help improve power availability and reduce dependence on conventional backup power systems for suitable networking applications.

5.3 RECOMMENDATIONS

Based on the design, implementation, and testing of the developed system, the following recommendations are made:

I. **Use of Higher-Capacity Batteries:** Future versions of the system should incorporate higher-capacity batteries to increase the duration for which networking devices can operate during periods of insufficient solar energy.

II. **Improved Solar Panel Capacity:** A higher-capacity photovoltaic panel may be used where greater power demand is expected. This would improve battery charging performance and support longer operating periods.

III. **Integration of MPPT Charge Control:** Future improvements may incorporate Maximum Power Point Tracking (MPPT) technology to improve the efficiency of solar energy extraction, particularly under varying sunlight conditions.

IV. **Improved Battery State-of-Charge Monitoring:** More advanced battery monitoring techniques should be implemented to provide more accurate estimation of the remaining battery capacity rather than relying mainly on battery terminal voltage.

V. **Enhanced IoT Connectivity:** Future versions could incorporate additional communication options and improved data storage so that system information can remain accessible even when the primary local Wi-Fi connection is unavailable.

VI. **Automatic Load Management:** An intelligent load-management system could be incorporated to prioritize important networking devices and disconnect less critical loads when the battery reaches a low state of charge.

VII. **Improved Protection Features:** Additional electrical and thermal protection mechanisms should be incorporated to further protect the battery, voltage regulators, microcontroller, and connected networking equipment against abnormal operating conditions.

VIII. **Larger-Scale Implementation:** Further development should investigate the application of the system to larger networking installations by increasing the solar generation, battery storage, and power-conversion capacity according to the required load.

IX. **Further Testing:** Future researchers should conduct longer-duration tests under different solar radiation levels and load conditions to further evaluate the reliability, efficiency, battery performance, and long-term stability of the system.

X. **Development of a More Advanced Monitoring Dashboard:** The IoT monitoring interface could be improved by incorporating historical data, graphical analysis, automated alerts, and performance reports to provide users with more detailed information about the system.

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APPENDIX A

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// LCD address and dimensions
LiquidCrystal_I2C lcd(0x27, 16, 2);

// Analog voltage measurement pins
const int BATTERY_PIN = A0;
const int OUTPUT_12V_PIN = A1;
const int OUTPUT_9V_PIN = A2;

// Indicator pins
const int GREEN_LED = 6;
const int YELLOW_LED = 7;
const int RED_LED = 8;
const int BUZZER_PIN = 9;

// Voltage measurement settings
const float ADC_REFERENCE = 5.00;
const float DIVIDER_RATIO = 5.00;
const float ADC_MAX = 1023.0;

// Battery thresholds for a nominal 12V lead-acid battery.
// These are configurable monitoring thresholds, not a
// universal state-of-charge measurement.
const float LOW_BATTERY = 11.8;
const float CRITICAL_BATTERY = 11.3;
const float FULL_BATTERY_INDICATION = 12.7;

// Timing
```

```
const unsigned long UPDATE_INTERVAL = 1000;
unsigned long previousUpdate = 0;

// Measured voltage values
float batteryVoltage = 0.0;
float output12V = 0.0;
float output9V = 0.0;


// Read an analog input and convert it to the original voltage.
float readVoltage(int pin) {
  const int samples = 20;
  unsigned long total = 0;

  // Discard an initial conversion after switching channels.
  analogRead(pin);

  for (int i = 0; i < samples; i++) {
    total += analogRead(pin);
    delay(5);
  }

  float averageADC = total / (float)samples;

  float measuredVoltage =
      (averageADC * ADC_REFERENCE / ADC_MAX) * DIVIDER_RATIO;

  return measuredVoltage;
}


// Update the LEDs and buzzer according to battery voltage.
```

```
void updateBatteryIndicators() {
  digitalWrite(GREEN_LED, LOW);
  digitalWrite(YELLOW_LED, LOW);
  digitalWrite(RED_LED, LOW);
  digitalWrite(BUZZER_PIN, LOW);

  if (batteryVoltage <= CRITICAL_BATTERY) {
    digitalWrite(RED_LED, HIGH);
    digitalWrite(BUZZER_PIN, HIGH);
  } else if (batteryVoltage <= LOW_BATTERY) {
    digitalWrite(YELLOW_LED, HIGH);
  } else {
    digitalWrite(GREEN_LED, HIGH);
  }
}
// Print a voltage value to the LCD using a fixed-width field.
void printVoltage(float voltage) {
  if (voltage < 10.0) {
    lcd.print(" ");
  }

  lcd.print(voltage, 2);
  lcd.print("V");
}
// Display the battery and output measurements.
void displayReadings() {
  lcd.clear();

  // First line: battery voltage and battery condition.
  lcd.setCursor(0, 0);
  lcd.print("BAT:");
  printVoltage(batteryVoltage);
```

```
  if (batteryVoltage <= CRITICAL_BATTERY) {
    lcd.print(" LOW!");
  } else if (batteryVoltage <= LOW_BATTERY) {
    lcd.print(" LOW ");
  }

  // Second line: 12V and 9V output readings.
  lcd.setCursor(0, 1);
  lcd.print("12:");
  printVoltage(output12V);
  lcd.print(" 9:");
  printVoltage(output9V);
}

// Display measurements in the Serial Monitor.
void displaySerialReadings() {
  Serial.println("--------------------------------");

  Serial.print("Battery Voltage: ");
  Serial.print(batteryVoltage, 2);
  Serial.println(" V");

  Serial.print("12V Output: ");
  Serial.print(output12V, 2);
  Serial.println(" V");

  Serial.print("9V Output: ");
  Serial.print(output9V, 2);
  Serial.println(" V");

  Serial.print("Battery Condition: ");
```

```
  if (batteryVoltage <= CRITICAL_BATTERY) {
    Serial.println("CRITICAL");
  } else if (batteryVoltage <= LOW_BATTERY) {
    Serial.println("LOW");
  } else if (batteryVoltage >= FULL_BATTERY_INDICATION) {
    Serial.println("HIGH VOLTAGE / CHARGED RANGE");
  } else {
    Serial.println("NORMAL");
  }

  Serial.println("-------------------------------");
}
void setup() {
  Serial.begin(9600);

  pinMode(GREEN_LED, OUTPUT);
  pinMode(YELLOW_LED, OUTPUT);
  pinMode(RED_LED, OUTPUT);
  pinMode(BUZZER_PIN, OUTPUT);

  pinMode(BATTERY_PIN, INPUT);
  pinMode(OUTPUT_12V_PIN, INPUT);
  pinMode(OUTPUT_9V_PIN, INPUT);

  lcd.init();
  lcd.backlight();

  lcd.setCursor(0, 0);
  lcd.print("SOLAR GENERATOR");

  lcd.setCursor(0, 1);
  lcd.print("Initializing...");
```

```
  delay(2000);

  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("System Ready");

  delay(1000);

  Serial.println("Solar Generator Monitoring System");
  Serial.println("Arduino Uno initialized.");
}

void loop() {
  unsigned long currentTime = millis();

  if (currentTime - previousUpdate >= UPDATE_INTERVAL) {
    previousUpdate = currentTime;

    batteryVoltage = readVoltage(BATTERY_PIN);
    output12V = readVoltage(OUTPUT_12V_PIN);
    output9V = readVoltage(OUTPUT_9V_PIN);

    updateBatteryIndicators();
    displayReadings();
    displaySerialReadings();
  }
}
```

APPENDIX B

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